

THE GEOMETRY OF REFUSAL: WHY POST-HOC SAFETY IS FRAGILE AND PRETRAINING-TIME SAFETY PERSISTS

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ABSTRACT

Post-hoc safety training (RLHF, DPO) is the dominant way to align large language models, yet jailbreaks (Zou et al., 2023b), fine-tuning attacks (Qi et al., 2024), and activation-space probes (Arditi et al., 2024) keep recovering the behaviors it was meant to remove. We give this fragility one geometric explanation and trace it to *when*, during pretraining, safety can take hold. We measure the safety update $\Delta = W_{\text{safe}} - W_{\text{base}}$ against the curvature of the model’s capabilities (the empirical Fisher of a capability loss). Post-hoc safety consistently lands in a *suppression* regime: Δ is nearly orthogonal to the capability directions, and its small in-subspace part concentrates on a few high-curvature ones. The update is *thin* but *sharp*, a refusal gate laid over intact capabilities rather than erasure of them. A kernel-immobility lemma explains why such an update can only mask a capability, not remove it, so a little benign fine-tuning restores it: 100 steps of benign fine-tuning collapse refusal on Qwen-2.5-7B and Llama-3-8B Instruct at preserved capability, a signature that replicates across five model families.

Following the account into pretraining, a 267-checkpoint sweep of OLMo-2-1B (OLMo et al., 2025) shows the substrate that safety engages emerging in a sharp transition between roughly 6B and 60B pretraining tokens. We then use the account constructively: models trained from scratch with safety co-training spread *continuously* across pretraining reach 87 to 98% refusal whose post-attack level holds at 84 to 91% at every scale, an erosion of 2 to 14 pp against 35 to 38 pp for post-hoc installs, at capability matched or better than an LM-only baseline and holding from 410M to 6.9B, whereas a compute-matched *windowed* schedule installs no lasting refusal. Persistence of the safety signal across pretraining, not its timing, is what buys attack robustness.

1 INTRODUCTION

Large language models are aligned *after* pretraining, not during it. The standard pipeline (pretrain, then RLHF or DPO on a preference dataset) assumes that any undesired behavior a base model can exhibit can later be removed by a targeted fine-tuning signal. A consistent body of recent empirical work undermines that assumption: a few hundred steps of fine-tuning on innocuous data undo RLHF-installed refusals (Qi et al., 2024), targeted optimization finds suffix strings that recover “refused” behaviors at inference time (Zou et al., 2023b; Wei et al., 2023), and linear probes on hidden states often find the “refused” content still present and linearly decodable (Arditi et al., 2024; Zou et al., 2023a). These are not isolated engineering failures but symptoms of one structural fact about how post-hoc safety training meets the loss landscape of an already-pretrained model. Establishing that is harder than it sounds, for two reasons that shape the whole paper. A safety install is a displacement in a billion-dimensional weight space, and whether it *removed* a capability or merely hid it cannot be read off behavior, since a model that refuses and a model that cannot answer look identical from the outside; that is what forces a geometric instrument. And any one aligned checkpoint confounds

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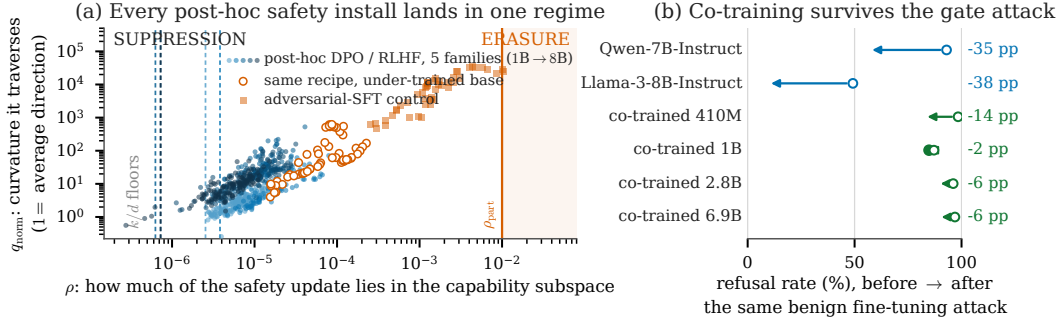


Figure 1: **Post-hoc safety lands orders of magnitude short of erasure and loses 35 to 38 pp of refusal to a benign fine-tuning attack; co-training holds 84 to 91%.** All points measured. **(a)** Where safety updates land in the (ρ, q_{norm}) plane, one point per transformer block and capability probe. Post-hoc DPO/RLHF across five model families (blue, light to dark by size) is *thin but sharp*: two to three orders of magnitude below the erasure boundary ρ_{part} , yet above each family’s own random-direction floor k/d (dashed; the five span $6.2\times$). Erasure is reachable but post-hoc safety never reaches it: an adversarial-SFT control built to saturate the capability subspace (squares) crosses ρ_{part} . Applying the same recipe to an under-trained base (circles) lands elsewhere, which Sec. 6 traces to a developmental transition. This panel is post-hoc installs only: a co-trained model never received a patch, so no such Δ exists and it is measured within-model instead (Sec. 7); its claim is panel (b). **(b)** Why that geometry matters: refusal rate before $\rightarrow$ after the *same* 100-example benign-SFT attack, regex-scored throughout so rows are comparable. Post-hoc installs lose 35 to 38 pp (Llama-3 to 11%); co-trained refusal ends at 84 to 91% at every scale to 6.9B. Regex tracks the refusal *gate*, which is what the geometry predicts; App. L repeats both attack tables under a harmful-output classifier.

the alignment recipe with the developmental stage of the base it was applied to, so separating them needs a pretraining checkpoint series, not a model.

Central claim. On a pretrained base with mature capabilities, post-hoc safety training moves weights nearly orthogonal to the principal capability directions; the small in-subspace component concentrates on high-curvature directions and functions as a thin refusal-gating circuit overlaid on an intact representational substrate (the internal features a refusal circuit attaches to; defined in Sec. 6), the geometric signature of a behavioral change without a representational one. The “mature” qualifier is load-bearing: before an identifiable developmental transition (Sec. 6), the same procedure lands elsewhere, at an order of magnitude higher curvature. We formalize this as a suppression/erasure dichotomy over (ρ, q_{norm}) , with two behaviorally distinguishable sub-patterns inside suppression, validated across 7 to 8B models. Figure 1 states the whole argument: every post-hoc install we measure falls in the same corner of the (ρ, q_{norm}) plane, and that is exactly the population the attack strips.

Contributions.

- **(C1) A weight-space classification** of safety installs into *suppression* ($\rho \ll \rho_{\text{part}}, q_{\text{norm}} \gg 1$) vs. *erasure* ($\rho \gtrsim \rho_{\text{part}}$), with two behaviorally distinct sub-patterns inside suppression. Validated across 5 model families (§4): ρ peaks at 21 to $66 \times k/d$ yet stays ≥ 2 orders below ρ_{part} , q_{norm} reaching 200+.
- **(C2) A kernel-immobility lemma** explaining why post-hoc safety masks rather than removes capability, with a stronger gradient-orthogonality form empirically tight at $|\cos(\nabla \mathcal{L}_{\text{cap}}, \Delta)| < 0.002$ (§5). Behaviorally, 100 benign Alpaca steps cut refusal to 58.2% (Qwen) and 10.9% (Llama) at preserved capability (§4).
- **(C3) A developmental signature**: substrate emergence between 1B and 100B pretraining tokens on OLMo-2-1B with cross-family Pythia-1.4B replication; per-revision DPO geometry transitions at the same point, with pre-substrate $q_{\text{norm}}^{\text{ARITH}} > 600$ (§6).
- **(C4) A constructive resolution**: from-scratch continuous safety co-training reaches **87 to 98%** pre-attack refusal that the benign supervised-fine-tuning (benign-SFT) attack leaves at 84 to

91% at matched-or-better capability, holding from 410M to 6.9B and robust to stronger attacks, while a compute-matched windowed control installs no refusal, isolating *persistence* of the safety signal (not its timing) as the driver (§7).

Reader’s guide. Two scalar diagnostics carry the whole paper: ρ , how much of the safety update overlaps the capability directions, and q_{norm} , how sharp those directions are, so a *suppression* install has ρ small and q_{norm} large (§3). Sections then take C1 to C4 in order.

2 RELATED WORK

Why post-hoc safety is fragile. RLHF and DPO (Christiano et al., 2017; Ouyang et al., 2022; Rafailov et al., 2023) are undone by benign fine-tuning (Qi et al., 2024; Zhan et al., 2024), adversarial suffixes (Zou et al., 2023b; Andriushchenko et al., 2025), and linear-probe recovery of refusal (Arditi et al., 2024; Wollschläger et al., 2025). The closest accounts localize the fragility: shallow alignment in the first output tokens (Qi et al., 2025), a few safety-critical parameters (Wei et al., 2024), and a narrow weight-space “safety basin” (Peng et al., 2024). Each finds a *symptom* of a thin install; our (ρ, q_{norm}) classification gives the continuous geometry that makes them one phenomenon, and Lemma 1 says why an install of that shape must be recoverable, which the mechanistic overlay-not-erasure results (Lee et al., 2024; Ardit et al., 2024; Hubinger et al., 2024) observe but do not explain.

When it can take hold. Critical periods in deep networks (Achille et al., 2018; Frankle et al., 2020; Kleinman et al., 2024; Constantinescu et al., 2025) and emergent abilities (Wei et al., 2022; Schaeffer et al., 2023) establish that training time matters; we show the substrate *safety* binds to has its own window. Fisher methods (Amari, 1998; Martens, 2020; Kirkpatrick et al., 2017; Grosse et al., 2023) and task arithmetic (Ilharco et al., 2023; Ortiz-Jiménez et al., 2023) supply the instrument; Pythia (Biderman et al., 2023) and OLMo-2 (OLMo et al., 2025; Lambert et al., 2024) the checkpoints.

Safety during pretraining. Closest to our constructive result, Korbak et al. (2023) inject preference feedback into the pretraining objective and lower toxicity, but give no geometric mechanism, no suppression-versus-erasure account, no developmental transition, and crucially no windowed-versus-continuous comparison: Sec. 7 isolates *persistence* of the safety signal, not its mere presence during pretraining, as what buys attack robustness. Unlearning (Li et al., 2024) targets the erasure regime instead, the contrast that anchors our dichotomy. App. N adds interpretability, Fisher methods, unlearning, and open-weight substrates.

3 SETUP AND GEOMETRIC INSTRUMENT

Models. Five families, each chosen for what it makes measurable. **Pythia** (Biderman et al., 2023) ships intermediate pretraining checkpoints, so Sec. 4 can apply our own DPO along a developmental axis at 1.4B; Sec. 7 retrains the architecture from scratch at 410M to 6.9B. **Qwen-2.5-7B** (Yang et al., 2024), **Meta-Llama-3-8B** (Grattafiori et al., 2024) and **Mistral-7B-v0.3** ship official Instruct variants, so W_{safe} is a real install and recoverability is well-defined (Sec. 4). **OLMo-2-0425-1B** (OLMo et al., 2025) is the substrate for Sec. 6: it is the one model shipping a dense checkpoint series across the whole of pretraining *and* an aligned Tulu-3 (Lambert et al., 2024) variant with real refusal behavior.

Safety training. When we train safety ourselves (Pythia-1.4B in Sec. 4, OLMo-2-1B in Sec. 6), we use DPO (Rafailov et al., 2023) on HH-RLHF (Bai et al., 2022a), 5K preference pairs, 1 epoch, $\beta = 0.1$, learning rate 5×10^{-7} . For the instruct-tuned 7 to 8B models (Sec. 4) and for the substrate sweep (Sec. 6) we use the shipped Instruct weights as W_{safe} without further training.

Capability probes. Four standard benchmarks, each operationalized as the negative log-likelihood of the gold answer span given the prompt: REASON (MMLU `formal_logic` (Hendrycks et al., 2020)), ARITH (GSM8K training split (Cobbe et al., 2021)), RECALL (TriviaQA RC-no-context (Joshi et al., 2017)), and INSTRUCT (FLAN-v2 (Longpre et al., 2023)). Throughout, “ca-

pability” means these general abilities (reasoning, arithmetic, recall, instruction-following), *not* the ability to produce harmful content. Our central claim is that post-hoc safety leaves the former intact while gating the latter’s expression; keeping the two senses distinct is essential to reading the paper, since “capability preserved” refers to general ability, not to harmful output.

Notation. Flattened parameters $\theta \in \mathbb{R}^d$ (geometric measurement uses per-layer MLP only, App. A). Pretraining trajectory $\{\theta_t\}_{t=0}^T$; a specific revision is $W_{\text{base}} = \theta_t$. Safety procedure $\mathcal{A}_{\text{safe}}$ maps W_{base} to W_{safe} ; $\Delta = W_{\text{safe}} - W_{\text{base}}$. For capability c with held-out dataset $\mathcal{D}_c$,

$$F_{\text{cap}}^{(c)} = \mathbb{E}_{x \sim \mathcal{D}_c} \left[\nabla_{\theta} \log p_{\theta}(y | x) \nabla_{\theta} \log p_{\theta}(y | x)^{\top} \right]_{\theta=W_{\text{base}}}.$$

We work with its top- k eigenspace $U_k^{(c)} \in \mathbb{R}^{d \times k}$, estimated from N per-example gradients via the N -side Gram trick.

The geometric instrument. Two scalars classify a safety install: ρ , how much of the update lies in the capability subspace, and q_{norm} , how sharp those in-subspace directions are. A *suppression* install is thin (ρ small) but sharp (q_{norm} large); an *erasure* install has ρ large.

Overlap: how much of the update is inside the capability subspace. Given Δ and $U_k^{(c)}$, decompose $\Delta = \Delta_{\parallel} + \Delta_{\perp}$ where $\Delta_{\parallel} = U_k^{(c)} (U_k^{(c)})^{\top} \Delta$. The *overlap ratio* is

$$\rho^{(c)} = \frac{\|\Delta_{\parallel}\|_2^2}{\|\Delta\|_2^2} \in [0, 1].$$

A random k -subspace in an ambient of dimension d captures k/d of a random direction’s squared norm, so meaningful Fisher alignment has $\rho \gg k/d$. Here $k = 128$ and d is the block’s MLP parameter count, so the floor is architecture-dependent (App. B, Table 4: full decomposition, and a test placing observed ρ at 157 to 517 standard deviations above it).

Curvature: how sharp the directions it traverses are. Overlap-free measurement of “how much does Δ actually move $\mathcal{L}_{\text{cap}}$ ” is

$$q^{(c)} = \frac{\Delta^{\top} F_{\text{cap}}^{(c)} \Delta}{\|\Delta\|_2^2}, \quad q_{\text{norm}}^{(c)} = \frac{q^{(c)}}{\text{tr}(F_{\text{cap}}^{(c)})/d}.$$

q_{norm} is scale-free: $q_{\text{norm}} = 1$ means Δ traverses average capability curvature; $q_{\text{norm}} \gg 1$ means Δ preferentially exploits high-curvature directions.

Per-layer measurement. The Fisher is block-diagonal-dominant in practice, so we report $\rho_{\ell}^{(c)}$ and $q_{\text{norm}_{\ell}}^{(c)}$ per transformer block ℓ , using the standard empirical Fisher (Amari, 1998; Martens, 2020; Sagun et al., 2017; Gur-Ari et al., 2018); Sec. 4 shows the per-layer curve itself carries structure.

Two regimes. Two reference scales fix the boundary: $\rho_{\text{rand}} = k/d$, the overlap a random direction gives, and $\rho_{\text{part}} = 10^{-2}$, where $\sim 10\%$ of $\|\Delta\|_2$ lies inside the capability subspace (anchored in App. G by an adversarial control that reaches exactly there). Below ρ_{part} is *suppression*, above it *erasure*, the target of unlearning (Li et al., 2024). Every DPO/RLHF install we measure is suppression, over two orders below ρ_{part} on a mature base and $44\times$ below on an under-trained one, and the split survives any cut in $[5 \times 10^{-3}, 2 \times 10^{-2}]$. Curvature reads separately: $q_{\text{norm}} \gg 1$ is a thin, sharp install, $q_{\text{norm}} \sim 1$ a diffuse one. Two sub-patterns sit inside suppression and separate behaviorally: an *under-trained base*, where the base has not converged, so safety training still lowers $\mathcal{L}_{\text{cap}}$ directly; and an *SFT-stage recipe*, where the supervised stage drives motion inside F_{cap} ’s top- k and leaves residual attack resistance. App. C gives both with their ρ and q_{norm} ranges.

4 POST-HOC SAFETY LIVES IN SUPPRESSION, AND SUPPRESSION PREDICTS FRAGILITY

On a fully pretrained base, post-hoc DPO lands uniformly in the suppression regime, ρ well above the random floor yet two to three orders of magnitude below the erasure boundary and capability loss

not rising, and that signature predicts behavioral fragility: on 7 to 8B instruct models, 100 benign fine-tuning steps halve or fully remove refusal at preserved general capability.

As a first check, the geometry probe on Pythia-1.4B (step143000, DPO’d on 5K HH-RLHF pairs) lands squarely in suppression: across all four probes and 24 blocks $\rho \in [5 \times 10^{-6}, 8 \times 10^{-5}]$ (2 to $20\times$ the random floor, two to three orders below ρ_{part}), per-probe $q_{\text{norm}\ell}$ reaches 21, and capability loss *decreases* on every probe, so nothing is erased (full table and per-layer figure in App. D). Pythia-1.4B produces no measurable refusal at this scale, so we test the behavioral half of the claim on instruct-tuned 7 to 8B models, where refusal is real and its removal measurable.

Qwen-2.5-7B’s shipped Δ raises refusal by 22 pp rather than creating it. We treat Qwen-2.5-7B as W_{base} and Qwen-2.5-7B-Instruct as W_{safe} : the instruct model refuses 93% of AdvBench prompts and the base already refuses 71% (Table 10).

The geometry is suppression: a thin high-leverage gate, not erasure. Across all four probes and 28 blocks, ρ runs 5 to $65\times$ the random-overlap floor yet stays 2 to 3 orders of magnitude below ρ_{part} , while q_{norm} reaches the low hundreds on REASON and ARITH and stays flat on INSTRUCT (per-probe values and the per-layer curves in App. E). The update is nearly orthogonal to the capability Fisher, and what little of it lies in-subspace concentrates on high-curvature reasoning directions.

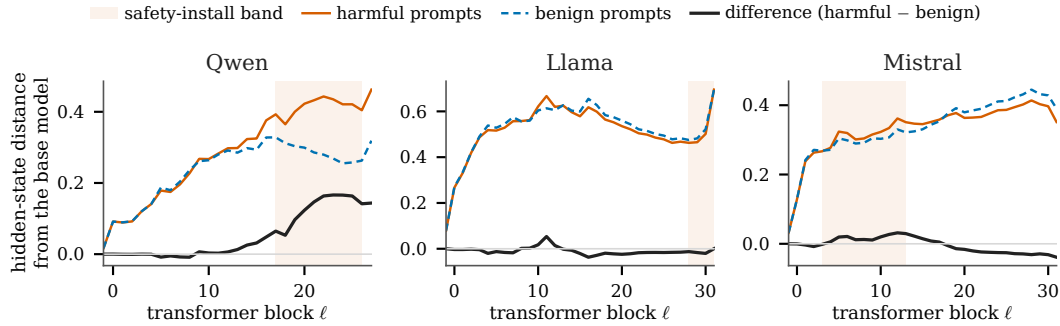


Figure 2: **What the install added is flat through the stack until it turns on inside a narrow band of blocks.** Per-layer L2 distance between base and safe residual streams over 50 harmful and 50 topic-matched benign prompts; the two curves track each other until late in the stack, and their *difference* (black) is what the install added, shaded where it turns on. That band is the *safety-install band*, and it coincides with where q_{norm} peaks (App. E, Fig. 5) and with the refusal direction reported by Arditi et al. (2024): three independent measurements agree on the layer. Its position is family-specific, mid-stack on Qwen and Mistral, at the very top on Llama.

Hidden-state divergence is layer-localized (Fig. 2). For each of 50 harmful and 50 benign topic-matched prompts we measure the per-layer L2 distance between base and safe residual streams. Selective divergence (harm – benign) sits at noise through the lower two thirds of the stack and then grows sharply, so instruct-tuning’s behavior-relevant changes concentrate in a band we call the *safety-install band*.

100 benign fine-tuning examples cost Qwen 34.8 pp of AdvBench refusal (Fig. 1b, top row). We fine-tune Qwen-2.5-7B-Instruct for 200 steps at $\text{lr}=2 \times 10^{-5}$ on 100 benign Alpaca examples and re-evaluate on AdvBench, HarmBench and StrongReject; Table 1 gives all three models on all three benchmarks. We report refusal rather than ASR because the two are exact complements under the HarmBench convention (Mazeika et al., 2024; Qi et al., 2024). None of the 100 examples demonstrates harmful content, which is what the suppression reading predicts: refusal is installed in a thin, high-curvature slice of weight space that benign SFT perturbs as a side effect, while the underlying capability, never erased per $\rho \ll k/d$, immediately routes to the original output distribution.

Capability-loss shift is a format artifact. $\Delta\mathcal{L}_{\text{cap}}$ is uniformly positive on raw 0-shot prompts, up to +3.35 on REASON, which might suggest partial erasure. A four-condition format ablation

(Appendix F) rules this out: prepending three in-context demonstrations collapses the gap by 40 to 74%. Raw capability is preserved, as $\rho \ll k/d$ already predicts; the benign-SFT attack recovers exactly this preserved capability.

Table 1: **The attack takes 35 to 38 pp of refusal from both models that had any, leaving Llama-3 at 11%.** Refusal rate (%) before $\rightarrow$ after the same 100-example benign-SFT attack on the three shipped instruct models. Unlike Table 3 the columns carry no in- or out-of-distribution label: these models’ alignment data is not public, so which benchmark is in-distribution for them is unknown. Mistral ships with almost no refusal, so its row measures install-thinness. $n = 256/100/64$; the one cell that rises is within noise. App. L repeats the table under a harmful-output classifier.

Model	AdvBench	HarmBench	StrongReject
Qwen-2.5-7B	93.0 $\rightarrow$ 58.2	67.5 $\rightarrow$ 45.5	52.3 $\rightarrow$ 56.6
Llama-3-8B	49.2 $\rightarrow$ 10.9	46.5 $\rightarrow$ 6.5	46.5 $\rightarrow$ 7.8
Mistral-7B-v0.3	1.2 $\rightarrow$ 0.4	1.5 $\rightarrow$ 0.0	3.5 $\rightarrow$ 0.8

Llama-3 and Mistral replicate the regime, and the attack strips nearly all of Llama-3’s refusal. Running the full pipeline on both reproduces suppression with family-specific q_{norm} bands (App. E). Behaviorally the attack removes Llama-3-Instruct’s refusal on every benchmark, while Mistral ships almost none to begin with (Table 1). As an instrument-validation check, DPO on Pythia-1.4B (about 0% refusal) leaves the divergence instrument at noise across all layers, confirming it tracks exercised refusal behavior rather than weight motion (App. D).

5 KERNEL IMMOBILITY: POST-HOC SAFETY MASKS CAPABILITY BUT CANNOT REMOVE IT

A kernel-immobility lemma explains what Sec. 4 just measured: a safety update confined to the near-kernel of the capability Fisher can mask a capability’s output but cannot reduce the capability itself at first order, which is why benign fine-tuning recovers it. When the safety update is nearly orthogonal to the capability Fisher ($\rho \ll \rho_{\text{part}}$) and the base has converged on the capability, the lemma below makes this precise: no first-order post-hoc update reduces the capability at leading order.

Lemma 1 (Kernel immobility). *If $\Delta \in \ker(F_{\text{cap}}^{(c)})$ and $\mathcal{L}_{\text{cap}}^{(c)}$ has Lipschitz Hessian, then $\mathcal{L}_{\text{cap}}^{(c)}(W_{\text{base}} + \Delta) - \mathcal{L}_{\text{cap}}^{(c)}(W_{\text{base}}) = \frac{1}{2} \Delta^\top E \Delta + O(\|\Delta\|^3)$, with $E := H - F_{\text{cap}}^{(c)}$ the Gauss-Newton residual. Under GN ($E = 0$ at a well-specified mature checkpoint (Martens, 2020)) this is $O(\|\Delta\|^3)$. No first-order post-hoc update reduces $\mathcal{L}_{\text{cap}}^{(c)}$ at leading order. (Proof in App. H.)*

The lemma makes the §4 observation crisp: in-kernel Δ cannot erase the capability, only mask its output expression, the asymmetry of capability preservation with refusal change (Qi et al., 2024; Arditi et al., 2024). The kernel condition alone forces $\Delta^\top \nabla \mathcal{L}_{\text{cap}} = 0$, so the lemma applies even where W_{base} is not a stationary point of $\mathcal{L}_{\text{cap}}$; the local-minimum / well-specification assumption enters only at the Gauss-Newton step. App. I measures $\|\nabla \mathcal{L}_{\text{cap}}\|$ across the OLMo and Pythia developmental axes, a U-shape minimized in the compute-optimal band, large on both under-trained and over-trained sides ($\sim 200\times$ -Chinchilla on OLMo, replicated on Pythia at $\sim 5\times$ -Chinchilla), but $|\cos(\nabla \mathcal{L}_{\text{cap}}, \Delta)| < 0.002$ at every revision including the over-trained 4001B endpoint, consistent with kernel-induced gradient orthogonality. Frontier-scale models near Chinchilla (e.g., Llama-3-8B at ~ 75 tokens/param) sit where the GN residual $\|E\|$ is also small.

The orthogonality is not an artifact of the mixed update. The shipped $\Delta = W_{\text{safe}} - W_{\text{base}}$ folds instruction-following and safety together, so its near-orthogonality to $\nabla \mathcal{L}_{\text{cap}}$ could in principle come from the instruction-following part rather than from safety. To isolate safety, we train two *controlled* safety-only updates on a fixed base (Qwen-2.5-1.5B) with no general instruction-tuning: a refusal-SFT-only update and a DPO-only update. Both reproduce the orthogonality: $|\cos(\nabla \mathcal{L}_{\text{cap}}, \Delta)| \leq 2 \times 10^{-4}$ on every capability probe (reasoning, arithmetic, recall, instruction-following), for both recipes. The suppression signature is therefore a property of the safety update itself, not a confound of the mixed instruction-plus-safety delta.

6 THE SAFETY SUBSTRATE EMERGES IN A SHARP PRETRAINING WINDOW

The representational substrate that a refusal install engages emerges in a sharp transition between roughly 6B and 60B pretraining tokens; before it, the same safety procedure lands in a different geometric regime.

By *representational substrate* we mean the internal features a final-stage refusal circuit must attach to in order to gate a capability: the mid-to-late-layer representations that separate harmful from benign inputs. Before these exist, a safety install has nothing to engage. The *developmental transition* is the point in pretraining at which they appear, which we now locate.

Section 4 described post-hoc alignment at *mature* pretraining stages; we now ask when that regime-behavior emerges during pretraining. Two complementary sweeps answer it: one holds a fixed final-stage safety install and sweeps the base across 9 pretraining revisions (§6), the other applies the same DPO recipe at each revision and asks whether the weight-space geometry itself shifts (§6).

Why the sweep needs OLMo-2-0425-1B. Pythia-1.4B cannot support a developmental-axis study directly: our DPO recipe produces zero pre-attack refusal rate at any Pythia revision we tested (step 512, 30k, 143k), so any per-revision behavioral signal is vacuous. We therefore move to OLMo-2-0425-1B (OLMo et al., 2025), which ships 267 pretraining checkpoints (step 0 through step 1.9×10^6 , 0 to 4T tokens) and an already-aligned Instruct variant trained by AI2 with the Tulu-3 recipe (Lambert et al., 2024). We exploit the Instruct model’s real refusal behavior: we do not re-train safety ourselves in the substrate sweep.

Substrate question. Holding $W_{\text{safe}} = \text{OLMo-2-0425-1B-Instruct}$ fixed, we sweep W_{base} across 9 pretraining revisions spanning 0 to 4T tokens (Table 13). At each revision we measure the per-layer selective hidden-state divergence $\text{selective_div}_\ell^{(t)} = \text{harm_div}_\ell - \text{benign_div}_\ell$ (same 50-harm, 50-benign topic-matched prompt instrument as Sec. 4). This measures: *does the base model at pretraining time t have the representational substrate for the final-stage Instruct weights to differentially engage on harmful prompts?*

The substrate appears in a sharp band between 6B and 60B tokens, not gradually. Selective divergence sits at noise through the first 1B tokens, rises by two orders of magnitude inside that window, and plateaus thereafter (Fig. 3; per-step values in Table 13). The peak-divergence layer ℓ^* also migrates forward in the residual stream as pretraining proceeds, from the upper-middle blocks at the transition to the lower-middle blocks once mature.

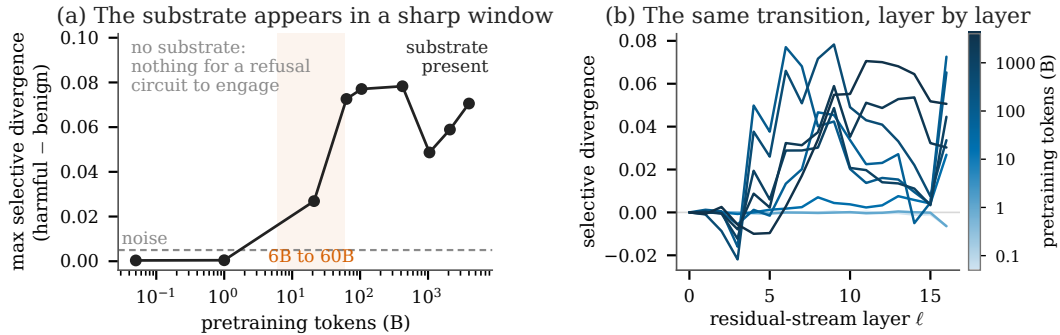


Figure 3: **Below roughly 6B pretraining tokens the base carries nothing for a refusal circuit to engage; by 60B the substrate is there.** OLMo-2-0425-1B substrate sweep, the final-stage Instruct weights held fixed while the base is swept across 9 pretraining revisions. **(a)** Max selective divergence (how much further the safety install moves the hidden state on harmful than on benign prompts) against pretraining tokens: at noise, then up two orders of magnitude, then a plateau. **(b)** The same measurement resolved per layer, one curve per revision, darker with more pretraining. Among post-substrate revisions the peak layer settles forward through the residual stream ($\ell^* = 6$ at 105B, 9 at 420B to 2T, 11 at 4T), having sat at the final block ($\ell^* = 16$) at the transition’s early edge.

The transition is where the base acquires features a refusal circuit can read. The sweep’s fixed- W_{safe} design does *not* measure “can DPO at time t install safety”, but the prior question just posed. The transition at $\sim 100\text{B}$ tokens is the point at which OLMo-2-0425-1B acquires that representational substrate. Before the transition, W_{safe} ’s safety weights have nothing to differentially engage on; after, they do. The forward migration of ℓ^* is consistent with circuits being installed in progressively later residual-stream positions as mid-block computation deepens.

ρ and q_{norm} stay flat through mid-training and spike at the final checkpoint. ρ declines through pretraining (1.4×10^{-6} at step 0, 8.3×10^{-7} at step 1M) and then jumps $4\times$ at the final checkpoint (3.4×10^{-6}). q_{norm} for ARITH and INSTRUCT tracks the same shape: flat (<5) through mid-training, spiking to $q_{\text{norm}_{\text{ARITH}}} = 19$, $q_{\text{norm}_{\text{INSTRUCT}}} = 27$ at step 1.9×10^6 . The spike is the “thin but sharp” signature from Sec. 4 appearing exactly when Δ is the pure instruct-tuning delta at the matched checkpoint, internally consistent with the regime observed on Qwen.

Per-revision DPO confirms the transition. Applying the *same* DPO recipe at revisions straddling the transition confirms the geometric shift is a property of the base’s representational stage, not of the alignment procedure, and exposes the two suppression sub-patterns of §3: an *under-trained-base* pattern, where q_{norm} is high before the substrate and falls an order of magnitude after, and an *SFT-stage recipe* pattern, where that stage keeps both scalars elevated at every revision. Behaviorally only SFT+DPO installs measurable refusal, and only the mature checkpoint retains any of it after the attack (Appendix J).

7 CONTINUOUS CO-TRAINING: PERSISTENCE, NOT TIMING, BUYS ROBUSTNESS

Injecting safety continuously through pretraining, so the signal persists to the end, installs refusal the benign-SFT attack largely fails to erode; a compute-matched early window installs none. Persistence, not timing, drives robustness. If post-hoc edits are trapped in suppression (Sec. 5) and the substrate they need appears only mid-pretraining (Sec. 6), safety should be injected during pretraining itself, so we ask which schedule $\lambda(t)$ in $\mathcal{L}_{\text{pre+safe}}(\theta; t) = \mathcal{L}_{\text{LM}}(\theta) + \lambda(t) \mathcal{L}_{\text{safe}}(\theta)$ yields refusal that survives the attack at matched capability.

Protocol. Four Pythia-410M models are trained from scratch on 10B C4 tokens: `base` (LM only); `cotrain-windowed` ($\lambda_0 = 0.5$ over $[0.10, 0.30]$, overlapping the OLMo substrate transition); `cotrain-continuous` ($\lambda_c = 0.10$ throughout, equal total safety weight); and `post-hoc-dpo` (5K-pair HH-RLHF DPO at the end). All face the benign-SFT attack of Sec. 4; full protocol in App. K.

Only `cotrain-continuous` installs refusal, at capability matched or better than base. It is the only one of the four that installs any (Table 18, with its attack numbers). `cotrain-windowed`’s 20% window cannot survive the 70% of LM-only training that follows, the wash-out also seen at 160M (App. K), and `post-hoc-dpo` installs nothing either.

Two limits on what that comparison can show. The failing conditions install no refusal, so there is no matched-refusal comparison: co-trained refusal survives, but we cannot show it survives *better* than an equally strong post-hoc install. And Δ has no counterpart here, since a co-trained model never received an edit; the only difference available is between two complete runs, which carries the language-modeling trajectory, not a safety displacement. Lemma 1 presumes such a displacement, so co-training falls outside its premise rather than contradicting it (the within-model instrument does apply, Table 20).

A safety window that stops at 30% of training installs nothing, though it starts earlier than one that works. To separate *when* safety is injected from *how long* it persists, we train four conditions at 1B that differ only in the pretraining window in which the safety loss is active (Table 2). Both schedules that run to the end of pretraining install refusal the attack leaves largely intact; the bounded window installs none, like the LM-only base. Onset time is therefore not what matters: persistence to the end is, which rules out a privileged early window.

Table 2: **Both schedules that persist to the end of pretraining survive the attack; the bounded early window installs nothing.** From-scratch co-training refusal (regex, %) on 64 AdvBench prompts, pre $\rightarrow$ post benign-SFT attack. Conditions differ only in the fractional pretraining window in which the safety loss is active. The 1B 87.5 $\rightarrow$ 90.6 change is a one-to-two-prompt swing within noise; Table 3 is the higher-powered 256-prompt rerun of `cotrain-continuous`.

Condition (safety window)	410M (pre $\rightarrow$ post)	1B (pre $\rightarrow$ post)
<code>cotrain-continuous</code> [0, 1]	84.4 $\rightarrow$ 84.4	87.5 $\rightarrow$ 90.6
<code>cotrain-post-substrate</code> [0.30, 1]	n/a	84.4 $\rightarrow$ 84.4
<code>cotrain-windowed</code> [0.10, 0.30]	0.0 $\rightarrow$ 1.6	3.1 $\rightarrow$ 4.7
base (LM only)	0.0 $\rightarrow$ 0.0	0.0 $\rightarrow$ 0.0

Scale: the install strengthens to 6.9B (Table 3). We repeat the `continuous-versus-base` comparison from scratch at 410M, 1B, 2.8B and 6.9B on 256 prompts. Post-attack AdvBench refusal holds at 84 to 91% at every scale, an erosion of 2 to 14 pp by regex against 35 to 38 for post-hoc installs (Fig. 1b), while the base installs none anywhere. The install therefore lands in the same place across a $17\times$ parameter range; 410M erodes most only because it starts highest.

Table 3: **Post-attack AdvBench refusal holds at 84 to 91% at every scale to 6.9B.** From-scratch `cotrain-continuous` refusal (%), pre $\rightarrow$ post benign-SFT attack, by scale and prompt distribution. The recipe’s refusal templates are AdvBench-like, so only AdvBench is in-distribution. The LM-only base stays at or below 24.6% everywhere. $n = 256/100/64$; changes under ~ 5 pp are within noise. Classifier columns: App. L.

Scale	AdvBench (in-dist.)	HarmBench (out)	StrongReject (out)
410M	98.4 $\rightarrow$ 84.0	58.0 $\rightarrow$ 34.5	8.2 $\rightarrow$ 7.4
1B	87.1 $\rightarrow$ 84.8	56.0 $\rightarrow$ 32.0	7.8 $\rightarrow$ 8.2
2.8B	96.1 $\rightarrow$ 90.2	45.5 $\rightarrow$ 31.5	8.6 $\rightarrow$ 8.2
6.9B	96.9 $\rightarrow$ 90.6	32.5 $\rightarrow$ 39.0	9.0 $\rightarrow$ 10.9

Coverage is in-distribution, and narrows with scale. Resisting erosion and covering the harmful-prompt space are separate axes, and the install wins only the first. Refusal resists erosion on all three benchmarks, but its *coverage* does not follow: while AdvBench holds near 97% at 6.9B, HarmBench falls from 58.0 to 32.5% over the same $17\times$ range and StrongReject never exceeds 10.9% at any scale. The recipe’s five templates fit their own prompt distribution, and a larger model fits them more tightly.

Co-trained safety is entangled with instruction-following, not with capability at large. A measurement that never differences two runs does compare co-trained against post-hoc models: project the safety gradient, taken *inside* one model, onto that same model’s capability eigenspace to give ρ_{safe} (definition in App. M). Small ρ_{safe} means refusal moves without disturbing capability, a thin gate; large means the two are entangled. Against `base` at the same scale the probes split, and in the same direction at every scale: ρ_{safe} rises on INSTRUCT (3.6, 1.3, 6.5, 2.1 $\times$) and falls on REASON (0.90, 0.68, 0.54, 0.74 $\times$), with RECALL and ARITH mixed (Table 20). That is why an instruction-tuning attack erodes it at all, and why the erosion stops short of removal. We compare only within a scale, since $\rho_{\text{safe}}/(k/d)$ grows with d in both arms. The install also survives stronger attacks: worst-case 75% under a learning-rate and step sweep, 100% under an identity-shift persona (App. K).

Conclusion. Post-hoc alignment lands in the suppression regime, and Lemma 1 shows such an update masks rather than removes capability, which is why benign fine-tuning recovers it. Co-training continuously through pretraining escapes that regime, holding 84 to 91% refusal under the attack against 35 to 38 pp lost post-hoc, at matched-or-better capability to 6.9B: *persistence*, not timing, drives robustness. Whether thinness is diagnostic of fragility or causal can be tested by penalizing q_{norm} post-hoc. Both bounds on that claim, absolute rather than matched robustness and in-distribution coverage, are stated above; App. M has the rest.

AI USE STATEMENT

We used generative AI tools for *implementation*: parts of the training, geometry-instrument, and analysis code were scaffolded with an AI coding assistant, then read, edited, and tested by the authors. We did not use generative AI tools for synthetic data generation, theoretical modeling, mathematical formulation, proof assistance, hypothesis refinement, methodology design, qualitative analysis, translation, data cleaning, or result interpretation; in particular, Lemma 1 and its proof (Appendix H) were formulated and proved by the authors. Additionally, we used generative AI tools for drafting and copy-editing prose.

We have reviewed all AI-assisted work. AI-scaffolded code was read and tested by the authors before use, and every number and figure in the paper is regenerated by the released pipeline from saved run artifacts rather than reported from a transcript. Every claim we attribute to prior work was checked by the authors against the primary source. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

ETHICS STATEMENT

This work studies why a public fine-tuning attack (Qi et al., 2024) removes safety alignment, and uses that understanding to build more attack-robust alignment; it introduces no new attack and no new harmful capability, and operators should read $\rho \ll k/d$ as a fine-tuning-robustness diagnostic of install thinness, not as an attack recipe. All harmful-prompt evaluations use public benchmarks (AdvBench, HarmBench), and we report only aggregate refusal and geometry statistics, never harmful completions. The pretraining-time co-training we advocate requires foundation-model training compute, which concentrates this form of alignment in large organizations; we note the tradeoff rather than endorse it. Our use of language models as a research aid is set out in the AI use statement above; all experiments, analyses, and claims are the authors’.

REPRODUCIBILITY STATEMENT

The geometry instrument (overlap ρ , normalized curvature q_{norm} , and the top- k Fisher-eigenspace estimator) is specified in Sec. 3 and Appendix B, including the Gram-trick estimator and its memory budget. All models are public (Pythia, Qwen-2.5, Llama-3, Mistral, OLMo-2); every safety-training and attack recipe with its hyperparameters is given in Sec. 3 and Appendix A; and the developmental axis uses the released OLMo-2 and Pythia intermediate pretraining checkpoints. Lemma 1 is proved in full in Appendix H, and the co-training protocol with per-condition tables is in Appendix K. The full code is included in the supplementary material, covering the geometry instrument, the safety-training and attack recipes, the co-training pipeline, and the analysis scripts that regenerate every table and figure in the paper.

REFERENCES

- Alessandro Achille, Matteo Rovere, and Stefano Soatto. Critical learning periods in deep networks. In *ICLR*, 2018.
- Shun-ichi Amari. Natural gradient works efficiently in learning. *Neural Computation*, 1998.
- Maksym Andriushchenko, Francesco Croce, and Nicolas Flammarion. Jailbreaking leading safety-aligned LLMs with simple adaptive attacks. In *ICLR*, 2025. arXiv:2404.02151.
- Cem Anil, Esin Durmus, Mrinank Sharma, et al. Many-shot jailbreaking. *NeurIPS*, 2024.
- Andy Arditi, Oscar Balcells Obeso, Aaqib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. Refusal in language models is mediated by a single direction. In *NeurIPS*, 2024.
- Yuntao Bai, Andy Jones, Kamal Ndousse, et al. Training a helpful and harmless assistant with reinforcement learning from human feedback. *arXiv preprint arXiv:2204.05862*, 2022a.
- Yuntao Bai, Saurav Kadavath, Sandipan Kundu, et al. Constitutional ai: Harmlessness from ai feedback. *arXiv preprint arXiv:2212.08073*, 2022b.

-
- Stella Biderman, Hailey Schoelkopf, Quentin Anthony, et al. Pythia: A suite for analyzing large language models across training and scaling. In *ICML*, 2023.
- Paul F Christiano, Jan Leike, Tom Brown, Miljan Martic, Shane Legg, and Dario Amodei. Deep reinforcement learning from human preferences. *NeurIPS*, 2017.
- Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, et al. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.
- Ionut Constantinescu, Tiago Pimentel, Ryan Cotterell, and Alex Warstadt. Investigating critical period effects in language acquisition through neural language models. *TACL*, 2025. arXiv:2407.19325.
- Jonathan Frankle, David J Schwab, and Ari S Morcos. The early phase of neural network training. In *ICLR*, 2020.
- Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.
- Ryan Greenblatt, Carson Denison, Benjamin Wright, et al. Alignment faking in large language models. *arXiv preprint arXiv:2412.14093*, 2024.
- Roger Grosse, Juhan Bae, Cem Anil, et al. Studying large language model generalization with influence functions. *arXiv preprint arXiv:2308.03296*, 2023.
- Guy Gur-Ari, Daniel A Roberts, and Ethan Dyer. Gradient descent happens in a tiny subspace. *arXiv preprint arXiv:1812.04754*, 2018.
- Dan Hendrycks, Collin Burns, Steven Basart, et al. Measuring massive multitask language understanding. *arXiv preprint arXiv:2009.03300*, 2020.
- Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, et al. Training compute-optimal large language models. In *NeurIPS*, 2022. arXiv:2203.15556.
- Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: low-rank adaptation of large language models. In *ICLR*, 2022. arXiv:2106.09685.
- Tiansheng Huang, Sihao Hu, Fatih Ilhan, Selim Furkan Tekin, and Ling Liu. Lisa: Lazy safety alignment for large language models against harmful fine-tuning attack. *NeurIPS*, 2024a.
- Tiansheng Huang, Sihao Hu, and Ling Liu. Vaccine: perturbation-aware alignment for large language models against harmful fine-tuning attack. *NeurIPS*, 2024b. arXiv:2402.01109.
- Evan Hubinger, Carson Denison, Jesse Mu, et al. Sleeper agents: training deceptive LLMs that persist through safety training. *arXiv preprint arXiv:2401.05566*, 2024.
- Gabriel Ilharco, Marco Tulio Ribeiro, Mitchell Wortsman, Ludwig Schmidt, Hannaneh Hajishirzi, and Ali Farhadi. Editing models with task arithmetic. In *ICLR*, 2023.
- Arthur Jacot, Franck Gabriel, and Clément Hongler. Neural tangent kernel: Convergence and generalization in neural networks. In *NeurIPS*, 2018.
- Mandar Joshi, Eunsol Choi, Daniel S Weld, and Luke Zettlemoyer. Triviaqa: A large scale distantly supervised challenge dataset for reading comprehension. In *ACL*, 2017.
- Nitish Shirish Keskar, Dheevatsa Mudigere, Jorge Nocedal, Mikhail Smelyanskiy, and Ping Tak Peter Tang. On large-batch training for deep learning: Generalization gap and sharp minima. In *ICLR*, 2017.
- James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, et al. Overcoming catastrophic forgetting in neural networks. *PNAS*, 2017.

-
- Michael Kleinman, Alessandro Achille, and Stefano Soatto. Critical learning periods emerge even in deep linear networks. *ICLR*, 2024.
- Tomasz Korbak, Kejian Shi, Angelica Chen, et al. Pretraining language models with human preferences. In *ICML*, 2023.
- Nathan Lambert, Jacob Morrison, Valentina Pyatkin, et al. Tülu 3: Pushing frontiers in open language model post-training. *arXiv preprint arXiv:2411.15124*, 2024.
- Andrew Lee, Xiaoyan Bai, Itamar Pres, Martin Wattenberg, Jonathan K Kummerfeld, and Rada Mihalcea. A mechanistic understanding of alignment algorithms: A case study on DPO and toxicity. *ICML*, 2024.
- Nathaniel Li, Alexander Pan, Anjali Gopal, et al. The WMDP benchmark: measuring and reducing malicious use with unlearning. In *ICML*, 2024. arXiv:2403.03218.
- Tom Lieberum, Senthooan Rajamanoharan, Arthur Conmy, Lewis Smith, Nicolas Sonnerat, Vikrant Varma, Janos Kramar, Anca Dragan, Rohin Shah, and Neel Nanda. Gemma scope: open sparse autoencoders everywhere all at once on Gemma 2. *arXiv preprint arXiv:2408.05147*, 2024.
- Shayne Longpre, Le Hou, Tu Vu, et al. The flan collection: Designing data and methods for effective instruction tuning. *ICML*, 2023.
- Aengus Lynch, Phillip Guo, Aidan Ewart, Stephen Casper, and Dylan Hadfield-Menell. Eight methods to evaluate robust unlearning in LLMs. *arXiv preprint arXiv:2402.16835*, 2024.
- Kaifeng Lyu, Haoyu Zhao, Xinran Gu, Dingli Yu, Anirudh Goyal, and Sanjeev Arora. Keeping LLMs aligned after fine-tuning: The crucial role of prompt templates. *NeurIPS*, 2024.
- Samuel Marks and Max Tegmark. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets. *COLM*, 2024.
- Samuel Marks, Can Rager, Eric J Michaud, Yonatan Belinkov, David Bau, and Aaron Mueller. Sparse feature circuits: discovering and editing interpretable causal graphs in language models. In *ICLR*, 2025. arXiv:2403.19647.
- James Martens. New insights and perspectives on the natural gradient method. *JMLR*, 2020.
- Mantas Mazeika, Long Phan, Xuwang Yin, et al. Harmbench: A standardized evaluation framework for automated red teaming and robust refusal. *arXiv preprint arXiv:2402.04249*, 2024.
- Eric J Michaud, Ziming Liu, Uzay Girit, and Max Tegmark. The quantization model of neural scaling. In *NeurIPS*, 2024. arXiv:2303.13506.
- Team OLMo et al. 2 OLMo 2 furious. *arXiv preprint arXiv:2501.00656*, 2025.
- Guillermo Ortiz-Jiménez, Alessandro Favero, and Pascal Frossard. Task arithmetic in the tangent space: improved editing of pre-trained models. In *NeurIPS*, 2023. arXiv:2305.12827.
- Long Ouyang, Jeffrey Wu, Xu Jiang, et al. Training language models to follow instructions with human feedback. *NeurIPS*, 2022.
- Nina Panickssery, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. Steering llama 2 via contrastive activation addition. *ACL*, 2024.
- Anselm Paulus, Arman Zharmagambetov, Chuan Guo, Brandon Amos, and Yuandong Tian. AdvPrompter: fast adaptive adversarial prompting for LLMs. In *ICLR*, 2025. arXiv:2404.16873.
- ShengYun Peng, Pin-Yu Chen, Matthew Hull, and Duen Horng Chau. Navigating the safety landscape: measuring risks in finetuning large language models. *NeurIPS*, 2024. arXiv:2405.17374.
- Xiangyu Qi, Yi Zeng, Tinghao Xie, Pin-Yu Chen, Ruoxi Jia, Prateek Mittal, and Peter Henderson. Fine-tuning aligned language models compromises safety, even when users do not intend to. In *ICLR*, 2024.

-
- Xiangyu Qi, Ashwinee Panda, Kaifeng Lyu, Xiao Ma, Subhrajit Roy, Ahmad Beirami, Prateek Mittal, and Peter Henderson. Safety alignment should be made more than just a few tokens deep. In *ICLR*, 2025. arXiv:2406.05946.
- Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D Manning, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. In *NeurIPS*, 2023.
- Domenic Rosati, Jan Wehner, Kai Williams, Lukasz Bartoszcze, David Atanasov, Robie Gonzales, Subhabrata Majumdar, Carsten Maple, Hassan Sajjad, and Frank Rudzicz. Representation noising: a defence mechanism against harmful finetuning. *NeurIPS*, 2024. arXiv:2405.14577.
- Levent Sagun, Utku Evci, V. Uğur Güney, Yann Dauphin, and Léon Bottou. Empirical analysis of the Hessian of over-parametrized neural networks. *arXiv preprint arXiv:1706.04454*, 2017.
- Rylan Schaeffer, Brando Miranda, and Sanmi Koyejo. Are emergent abilities of large language models a mirage? *NeurIPS*, 2023.
- Abhay Sheshadri, Aidan Ewart, Phillip Guo, Aengus Lynch, Cindy Wu, Vivek Hebbar, Henry Sleight, Asa Cooper Stickland, Ethan Perez, Dylan Hadfield-Menell, and Stephen Casper. Latent adversarial training improves robustness to persistent harmful behaviors in LLMs. *arXiv preprint arXiv:2407.15549*, 2024.
- Aarohi Srivastava et al. Beyond the imitation game: Quantifying and extrapolating the capabilities of language models. *TMLR*, 2023.
- Adly Templeton, Tom Conerly, Jonathan Marcus, et al. Scaling monosemanticity: Extracting interpretable features from claude 3 sonnet. *Transformer Circuits Thread*, 2024.
- Alexander Wei, Nika Haghtalab, and Jacob Steinhardt. Jailbroken: How does LLM safety training fail? *NeurIPS*, 2023.
- Boyi Wei, Kaixuan Huang, Yangsibo Huang, Tinghao Xie, Xiangyu Qi, Mengzhou Xia, Prateek Mittal, Mengdi Wang, and Peter Henderson. Assessing the brittleness of safety alignment via pruning and low-rank modifications. In *ICML*, 2024. arXiv:2402.05162.
- Jason Wei, Yi Tay, Rishi Bommasani, et al. Emergent abilities of large language models. In *TMLR*, 2022.
- Tom Wollschläger, Jannes Elstner, Simon Geisler, Vincent Cohen-Addad, Stephan Günnemann, and Johannes Gasteiger. The geometry of refusal in large language models: concept cones and representational independence. In *ICML*, 2025. arXiv:2502.17420.
- Mitchell Wortsman, Gabriel Ilharco, Samir Ya Gadre, et al. Model soups: averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. In *ICML*, 2022.
- Prateek Yadav, Derek Tam, Leshem Choshen, Colin Raffel, and Mohit Bansal. Ties-merging: Resolving interference when merging models. *NeurIPS*, 2023.
- An Yang, Baosong Yang, Beichen Zhang, et al. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.
- Qiusi Zhan, Richard Fang, Rohan Bindu, Akul Gupta, Tatsunori Hashimoto, and Daniel Kang. Removing RLHF protections in GPT-4 via fine-tuning. In *NAACL*, 2024. arXiv:2311.05553.
- Chunting Zhou, Pengfei Liu, Puxin Xu, et al. Lima: Less is more for alignment. *NeurIPS*, 2023.
- Andy Zou, Long Phan, Sarah Chen, et al. Representation engineering: A top-down approach to ai transparency. *arXiv preprint arXiv:2310.01405*, 2023a.
- Andy Zou, Zifan Wang, Nicholas Carlini, Milad Nasr, J Zico Kolter, and Matt Fredrikson. Universal and transferable adversarial attacks on aligned language models. *arXiv preprint arXiv:2307.15043*, 2023b.

A EXPERIMENTAL DETAILS AND REPRODUCIBILITY

Attack implementation. Our fine-tuning attack is the benign-SFT variant from (Qi et al., 2024): full-parameter AdamW updates on 100 Alpaca pairs for 200 steps at learning rate 2×10^{-5} , cross-entropy over target tokens only. Refusal detection is regex-based over the first 64 generated tokens against 14 markers ("i can't", "i cannot", "i won't", "i will not", "i'm sorry", "i am sorry", "sorry, but", "as an ai", "i am unable", "i'm unable", "refuse", "not able to", "cannot provide", "can't provide"). Compliance NLL is the negative log-likelihood of the token sequence " Sure, here is" appended after the harmful prompt, under greedy decoding conditioning.

Test set. 64 AdvBench prompts sampled with `random.Random(101)` from the public CSV at `llm-attacks/llm-attacks/data/advbench/harmful_behaviors.csv`. Attack training set: 100 Alpaca pairs sampled with `random.Random(201)` from `tatsu-lab/alpaca`. Divergence evaluation: 50 harmful and 50 benign topic-matched prompts from the same sources, seeded with `random.Random(101)` and `random.Random(103)`.

Capability probes. REASON: 64 MMLU `formal_logic` multiple-choice items; gold answer is the single letter A/B/C/D. ARITH: 64 GSM8K-train problems, target is the numeric answer as a suffix. RECALL: 64 TriviaQA-RC no-context questions, target is the gold answer span. INSTRUCT: 64 FLAN-v2 prompts, target is the gold completion. Capability loss is the negative log-likelihood of the gold span conditional on the prompt.

OLMo-2-0425-1B revision identifiers. Substrate sweep uses the Hugging-Face branches `stage1-step0-tokens0B`, `stage1-step300-tokens1B`, `stage1-step10000-tokens21B`, `stage1-step30000-tokens63B`, `stage1-step50000-tokens105B`, `stage1-step200000-tokens420B`, `stage1-step500000-tokens1049B`, `stage1-step1000000-tokens2098B`, `stage1-step1907359-tokens4001B`. All are `stage1` branches (pretraining); we avoid the `stage2-ingredient*-*` branches (midtraining annealing), which would conflate data-mixture changes with pretraining-step progress.

Geometry measurement. $N = 128$ per-example gradients per (probe, layer) for 7 to 8B models (reduced from 256 for Pythia-1.4B to fit memory); $k = 128$ subspace rank. Per-layer MLP parameters only (the attention-head and layer-norm contributions to F_{cap} are omitted for budget and would at most add low-curvature directions that cannot meaningfully change the suppression/erasure classification).

Compute budget. All experiments ran on a 6 to $8 \times$ B200 node. Approximate per-experiment wall times: full geometry + benign-SFT attack + representation probe + divergence pipeline on a 7 to 8B Instruct model ≈ 30 to 60 min per pair (Qwen, Llama, Mistral); OLMo-2-1B substrate sweep across 9 pretraining revisions ≈ 45 min; per-revision SFT+DPO on OLMo-2-1B at 5 revisions (20K SFT + 20K DPO $\times 2$ epochs each) ≈ 3 hr; Pythia-410M from-scratch continuous-co-training run (10B tokens, 2-rank DDP) ≈ 14 hr; Pythia-160M continued-training pilot ≈ 1 hr; developmental-axis follow-up checks (multi-seed substrate sweep + cross-family gradient-norm) ≈ 30 to 45 min through a 6-GPU pool. Total compute envelope for the experiments in this paper: ~ 80 B200-GPU-hours; preliminary or failed experiments not reported add a comparable amount.

Code and artifacts. All launcher scripts, per-experiment result bundles, and a single merged `combined_bundle.json` (9 experiments, 1.0 MiB) are in the anonymized repository.

B GEOMETRY INSTRUMENT: IMPLEMENTATION NOTES

Orthogonal decomposition of Δ (detail for Sec. 3). The overlap ratio in Sec. 3 uses $\Delta = \Delta_{\parallel} + \Delta_{\perp}$ with $\Delta_{\parallel} := U_k^{(c)}(U_k^{(c)})^{\top} \Delta$. Here $U_k^{(c)} \in \mathbb{R}^{d \times k}$ stacks the top- k eigenvectors of the empirical capability Fisher (equivalently, the top- k right singular vectors of the per-example gradient matrix G). Because $U_k^{(c)}$ has orthonormal columns, $U_k^{(c)}(U_k^{(c)})^{\top}$ is the orthogonal projector onto the top- k

Fisher eigenspace and the decomposition is orthogonal: $\Delta_{\parallel} \perp \Delta_{\perp}$ and $\|\Delta\|_2^2 = \|\Delta_{\parallel}\|_2^2 + \|\Delta_{\perp}\|_2^2$ (Pythagoras). Orthonormality also gives $\|\Delta_{\parallel}\|_2^2 = \|(U_k^{(c)})^\top \Delta\|_2^2$, i.e., the squared norm of the projection equals the squared norm of its k -dimensional coefficient vector. We use this second form exclusively: the d -vector $\Delta_{\parallel}$ is never materialized, and $\rho^{(c)}$ reduces to a $(N,)$ -sized intermediate (see the top- k eigenspace estimator below). $\Delta_{\perp}$ lives in directions of approximately zero Fisher curvature (outside the top- k eigenspace) and therefore does not contribute to first-order capability change, which is why only $\|\Delta_{\parallel}\|_2^2$ appears in $\rho^{(c)}$'s numerator.

Top- k eigenspace estimator. Our estimator of the top- k eigenspace of $F_{\text{cap}}^{(c)}$ from per-example gradients uses the N -side Gram trick. For N per-example gradients $G \in \mathbb{R}^{N \times d}$, the $d \times d$ Fisher is never formed; instead we compute the $N \times N$ Gram $GG^\top$, eigendecompose it, and recover the top- k right singular vectors of G as the capability eigenbasis. At $d \sim 2 \times 10^8$ (MLP parameter count for a 7B model) and $N \sim 128$, the full $fp32$ per-example gradient matrix $G \in \mathbb{R}^{N \times d}$ is ~ 100 GiB (the $N \times N$ Gram $GG^\top$ itself is only ~ 65 KB). We chunk the Fisher matmul along d in 4M-element slices and accumulate in $fp32$, bounding peak GPU memory at ~ 2 GiB beyond the (N, d) gradient buffer. We never materialize the (d, k) capability-subspace matrix U_k explicitly: the projection $U_k^\top \Delta$ is computed directly from G , V_k , and singular values, which reduces to an $(N,)$ -sized intermediate. This is what makes the pipeline run at Qwen/Llama scale on a single GPU.

Significance of the ρ alignment (random-subspace null). The claim $\rho \gg k/d$ is a statement about a null: a random k -subspace overlaps the capability eigenspace by k/d in expectation. We quantify how far the observed alignment sits from that null by drawing random k -subspaces at the true parameter dimensionality and forming a z -score, $z = (\rho_{\text{obs}} - \text{mean}_{\text{null}}) / \text{std}_{\text{null}}$. Table 4 reports it for the three large models. The observed overlap is 21 to 66 times the k/d floor and 157 to 517 standard deviations above the random-subspace mean ($p \approx 0$, two-sided), so the alignment is small in absolute terms yet unambiguously non-random. The companion orthogonality statistic $|\cos(\nabla \mathcal{L}_{\text{cap}}, \Delta)| \approx 0.002$ sits about 150 standard deviations from a zero-mean null while still leaving Δ 99.8% orthogonal to the capability gradient; the lemma’s first-order conclusion ($\Delta \mathcal{L}_{\text{cap}} \leq 0$) we verify directly rather than through this statistic.

Table 4: Random-subspace significance of the Fisher alignment. ρ_{obs} is the observed overlap (max over layers); the null is a random k -subspace at the model’s MLP parameter dimensionality d ; $z = (\rho_{\text{obs}} - k/d) / \text{std}_{\text{null}}$. All $p \approx 0$ (two-sided). The alignment is 21 to 66 times the floor and hundreds of null standard deviations above it, so it is small but not a high-dimensional coincidence.

Model	ρ_{obs}	$\rho_{\text{obs}} / (k/d)$	z (SDs above null)
Qwen-2.5-7B	4.0×10^{-5}	$66\times$	517
Mistral-7B	3.2×10^{-5}	$44\times$	344
Llama-3-8B	1.5×10^{-5}	$21\times$	157

Prompt format for the per-example gradients. All ρ and q_{norm} values reported in the paper use the raw 0-shot prompt convention described in Appendix F (“Question: ... \nAnswer:” with no chat template and no in-context demonstrations), the same format used for L_{cap} . Each per-example gradient is $\nabla_{\theta} \sum_t \log p_{\theta}(y_t \mid x, y_{<t})$ for the full gold span $y = (y_1, \dots, y_T)$, a single token for REASON, multiple tokens for the other probes, evaluated at $\theta = W_{\text{base}}$. The top- k Fisher subspace therefore contains a mixture of (i) reasoning-relevant directions, which vary per example, and (ii) at most a handful of format-bias directions, which are shared across examples. With $k = 128$ and $N = 128$ examples, reasoning-relevant variance dominates: format-bias contributes at most a few of the k eigenvectors. Evaluating the Fisher at W_{base} (the completion-style base) rather than at W_{safe} further biases the top- k basis toward reasoning directions, because the base emits gold tokens natively and its log-likelihood’s sharpest directions are content-related, not format-related.

3-shot robustness check. We re-ran the full geometry measurement (all three model families, all four probes, all 28 to 32 layers) with three in-context demonstrations prepended to each probe prompt, to test whether the ρ values are sensitive to prompt format. Table 5 reports mean ρ across layers and peak- q_{norm} -layer, for the raw 0-shot (paper default) and raw 3-shot conditions.

Primary claim (regime classification). Mean ρ is essentially unchanged between 0-shot and 3-shot: within $\pm 5\%$ for Qwen and Llama on every probe, within $\pm 20\%$ for Mistral. All three models stay at least two orders of magnitude below $\rho_{\text{part}} = 10^{-2}$ under 3-shot. The suppression classification does not depend on prompt format.

Secondary claim (q_{norm} -peak layer). Peak- q_{norm} layer is reproducible on the diagnostic probe (REASON) for Llama (L30 $\rightarrow$ L30) and Mistral (L12 $\rightarrow$ L12), confirming the Fig. 5 shaded bands. Qwen’s REASON peak sits at L0 in both conditions, an embedding-layer artifact where raw-gradient magnitude is large but the q_{norm} mass relevant to our claim lives in the mid-layer secondary peak shown in Fig. 5. Peak- q_{norm} layer for ARITH, RECALL, INSTRUCT is less stable across prompt formats, but these probes’ peaks are not the basis of our safety-install-band argument.

The framework’s prediction, reasoning variance dominates the top- k subspace regardless of in-context format, so ρ and the regime label carry over, is confirmed.

Table 5: Geometry robustness check under raw 3-shot probe prompts. Mean ρ averaged across all transformer blocks; q_{norm} -peak reports the layer where $q_{\text{norm}}^{(c)}$ attains its maximum. All three families remain in the suppression regime ($\rho \ll \rho_{\text{part}} = 10^{-2}$) under 3-shot; peak- q_{norm} layer is stable on the probes driving the safety-install-band claim.

Model	Probe	mean ρ		q_{norm} -peak layer	
		0-shot	3-shot	0-shot	3-shot
Qwen-2.5-7B	REASON	1.40×10^{-5}	1.40×10^{-5}	L0	L0
	ARITH	1.21×10^{-5}	1.01×10^{-5}	L26	L0
	RECALL	1.20×10^{-5}	9.37×10^{-6}	L18	L27
	INSTRUCT	1.07×10^{-5}	1.05×10^{-5}	L27	L17
Llama-3-8B	REASON	5.07×10^{-6}	5.07×10^{-6}	L30	L30
	ARITH	6.08×10^{-6}	7.02×10^{-6}	L30	L30
	RECALL	5.81×10^{-6}	5.87×10^{-6}	L13	L31
	INSTRUCT	5.38×10^{-6}	5.41×10^{-6}	L31	L31
Mistral-7B-v0.3	REASON	1.21×10^{-5}	1.21×10^{-5}	L12	L12
	ARITH	1.34×10^{-5}	1.30×10^{-5}	L4	L29
	RECALL	9.42×10^{-6}	7.64×10^{-6}	L31	L0
	INSTRUCT	1.16×10^{-5}	1.12×10^{-5}	L3	L12

C REGIME CLASSIFICATION SUMMARY

Table 6 summarizes the dichotomy introduced prose-form in §3: *suppression* ($\rho \ll \rho_{\text{part}}$, Lemma 1 applies whenever the base is mature) vs. *erasure* ($\rho \gtrsim \rho_{\text{part}}$, Lemma fails; unobserved under DPO/RLHF; reachable by adversarial SFT, App. G; target of unlearning (Li et al., 2024)). Every DPO/RLHF observation in this paper lies in the suppression row, our largest ρ is $\approx 2 \times 10^{-4}$, at least $50\times$ below $\rho_{\text{part}} = 10^{-2}$. We identify two behaviorally-distinguishable sub-patterns *within* suppression, detailed in the right column.

Behavioral implication per sub-pattern. The suppression baseline gives the “capability preserved, refusal fragile” asymmetry demonstrated in §4: the Qi benign-fine-tuning attack removes the thin gating circuit without touching the (off-subspace) capability representations. Sub-pattern (a) means DPO can reduce $\mathcal{L}_{\text{cap}}$ at first order (Lemma assumption fails) and therefore installs a weaker, already-moving refusal; we observe $q_{\text{norm}}^{\text{ARITH}}$ up to 863 at pre-substrate OLMo step 300 and behaviorally, under SFT+DPO, a 21.9% pre-attack refusal that the attack collapses to 0%. Sub-pattern (b) installs a refusal whose in-subspace SFT component provides residual resistance: the only per-revision run that retains non-zero post-attack refusal (15.6%) is SFT+DPO at the final OLMo checkpoint. Erasure would decouple the capability from the model at leading order; the Lemma shows standard post-hoc alignment cannot reach it.

Table 6: Regime dichotomy with two sub-patterns inside suppression. Every DPO/RLHF recipe we tested (Pythia-1.4B, Qwen-2.5-7B, Llama-3-8B, Mistral-7B-v0.3, OLMo-2-1B under Weak / Strong / SFT+DPO) lands in row 1; row 2 is unobserved under post-hoc alignment and is the target of unlearning (Li et al., 2024).

Regime	ρ	q_{norm}	Observed-in; sub-patterns within
Suppression	$\ll \rho_{\text{part}}$ (10^{-6} to 10^{-4} observed)	$\gg 1$ (thin and sharp) or ~ 1 (diffuse)	Baseline (pure DPO/RLHF on mature base, $\rho \sim 10^{-6}$ to 10^{-5}): Pythia-1.4B+DPO (§4); Qwen/Llama/Mistral-Instruct (§4); post-substrate OLMo Weak/Strong (§6). Sub-pattern (a), under-trained base: Lemma assumption fails, $q_{\text{norm}}^{\text{ARITH}} \in [600, 863]$ (pre-substrate OLMo step 300, §6). Sub-pattern (b), SFT-stage recipe: SFT’s gradient lies in F_{cap} ’s top- k , raising ρ 10 to $40\times$ to $\sim 10^{-4}$ (SFT+DPO every revision, §6).
Erasure	$\gtrsim \rho_{\text{part}}$	any	unobserved under DPO/RLHF; reachable by adversarial SFT on (prompt, wrong-answer) pairs (max $\rho \approx 10^{-2}$ on target probe; App. G); target of unlearning (RMU and related) (Li et al., 2024)

Table 7: Pythia-1.4B post-DPO: per-probe ρ (mean over 24 transformer blocks), normalized Fisher-quadratic form q_{norm} , and capability-loss shift $\Delta\mathcal{L}_{\text{cap}}$. All rows are in the suppression regime; capability loss *decreases* slightly on every probe.

Probe	mean ρ_{ℓ}	mean $q_{\text{norm}\ell}$	$\Delta\mathcal{L}_{\text{cap}}$	Regime
Reason (MMLU fl)	1.7×10^{-5}	7.30	−0.015	suppression
Arith (GSM8K)	2.4×10^{-5}	21.39	−0.027	suppression
Recall (TriviaQA)	2.2×10^{-5}	10.73	−0.010	suppression
Instruct (FLAN)	2.0×10^{-5}	6.50	−0.007	suppression

D PYTHIA-1.4B: FULL TABLES AND PER-LAYER FIGURE

Null behavior: the recipe installs no refusal. DPO on Pythia-1.4B step143000 with our recipe produces a pre-attack refusal rate of 0% on AdvBench. ASR and activation-probe AUC are therefore dominated by probe saturation rather than by meaningful alignment properties. Applying the same topic-matched selective-divergence instrument as in Fig. 5 (50 harmful, 50 benign), $|\text{selective_div}_{\ell}| < 10^{-3}$ at every layer, with no monotone structure. For reference, the Qwen-7B-Instruct curve on the same instrument reaches +0.166. The Pythia null is therefore an internal consistency check: when no refusal circuit is installed behaviorally, the representation-shift instrument sees nothing. This is what motivated the move to OLMo-2-1B-Instruct (which ships with a Tulu-3-trained install that does produce refusal) for the substrate sweep (Sec. 6).

E CROSS-FAMILY REPLICATION DETAILS

Learned-classifier scoring. The same generations rescored by a harmful-output classifier are in App. L, which repeats both main-text attack tables under both detectors.

Table 8: Qwen-2.5-7B $\rightarrow$ Qwen-2.5-7B-Instruct: per-probe overlap ρ (mean / max over 28 blocks), normalized Fisher-quadratic form q_{norm} , raw-0-shot $\Delta\mathcal{L}_{\text{cap}}$. Source for the prose summary in §4.

Probe	mean ρ	max ρ	mean q_{norm}	max q_{norm} (layer)	$\Delta\mathcal{L}_{\text{cap}}$	Regime
Reason	1.4×10^{-5}	4.1×10^{-5}	76.2	220.6 ($\ell=19$)	+3.45	suppression
Arith	1.2×10^{-5}	2.9×10^{-5}	47.1	198.4 ($\ell=26$)	+3.15	suppression
Recall	1.2×10^{-5}	2.2×10^{-5}	29.8	94.7 ($\ell=18$)	+1.07	suppression
Instruct	1.1×10^{-5}	2.1×10^{-5}	18.6	31.3 ($\ell=27$)	+0.67	suppression

Overlap stays far below the erasure boundary at every layer and every probe (Pythia-1.4B after DPO)

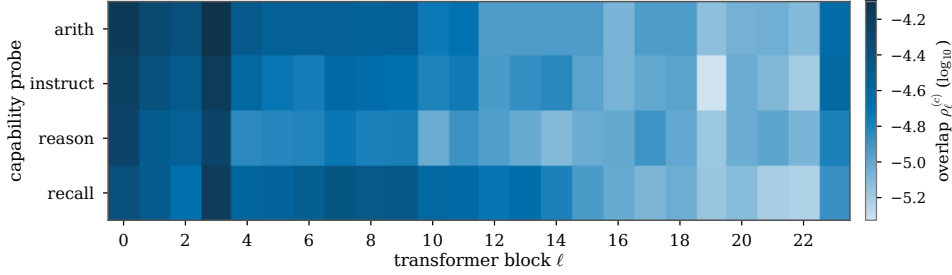


Figure 4: Per-layer ρ_ℓ (log-scale) for each capability probe on Pythia-1.4B after DPO. Layers 0, 3, and 23 (residual-stream boundaries and the final block) dominate; middle layers (~ 16 to 22) sit near the random-subspace floor. Pattern is consistent with per-layer localization reported for the refusal direction (Arditi et al., 2024).

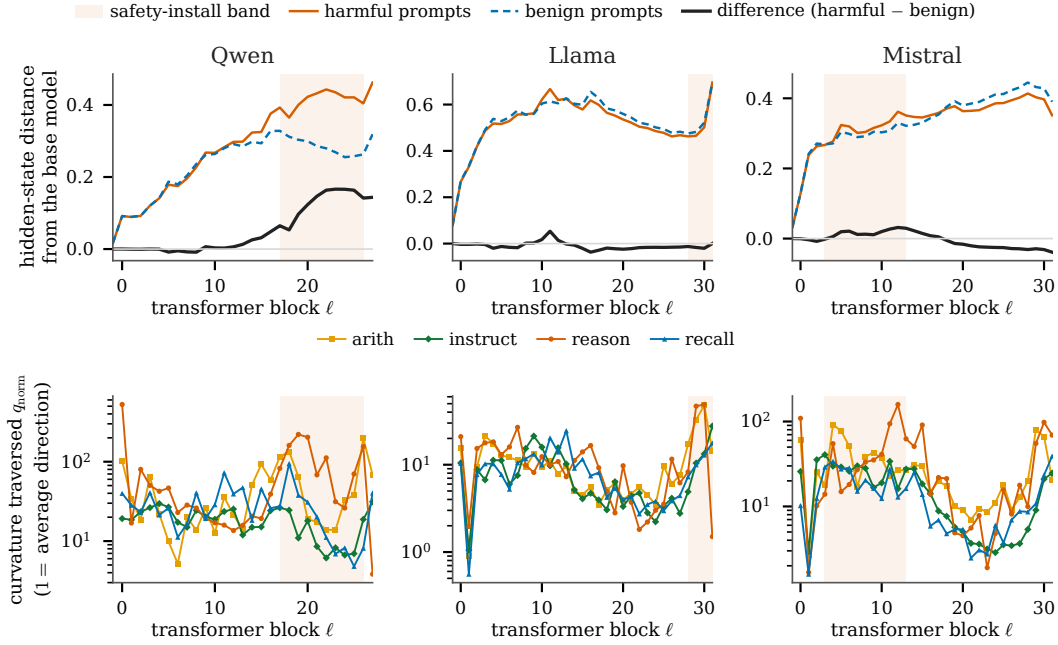


Figure 5: Layer-localized signals across three model families (columns: Qwen-2.5-7B-Instruct, Meta-Llama-3-8B-Instruct, Mistral-7B-Instruct-v0.3), referenced from Sec. 4. The top row is Fig. 2 in the main text, repeated here so the two rows can be read against each other. **Top row:** per-layer L2 divergence between base and instruct residual streams on harmful vs. benign topic-matched prompts ($n_{\text{pairs}}=50$ each), plus selective divergence (harm – benign). **Bottom row:** per-layer $q_{\text{norm}}^{(c)}$ for the four capability probes (log-scale). Shaded bands bracket each model’s $q_{\text{norm}}^{\text{REASON}}/q_{\text{norm}}^{\text{ARITH}}$ peaks (Qwen $\ell=17$ to 26; Llama $\ell=28$ to 31; Mistral $\ell=3$ to 13). Selective-divergence magnitude scales with pre-attack refusal rate: +0.166 (Qwen, 93%), +0.053 (Llama, 49%), +0.032 (Mistral, 1.2%).

Llama-3-8B. We use Meta-Llama-3-8B as W_{base} and Meta-Llama-3-8B-Instruct as W_{safe} . $\rho \in [3 \times 10^{-6}, 1.5 \times 10^{-5}]$ across 32 blocks and four probes, 10 to $30 \times$ the random floor $k/d \approx 5 \times 10^{-7}$. q_{norm} peaks for REASON and ARITH at blocks 30 to 31 (48.9 and 47.7); much later than Qwen’s mid-block peak. The 100-example benign Alpaca fine-tune strips nearly all of its refusal (Table 1; compliance NLL 6.18 $\rightarrow$ 4.30). Selective divergence peaks at +0.053 at block 12, roughly $3 \times$ smaller than Qwen’s +0.166, with late layers going slightly negative (−0.02), reflecting that only 49% of harmful prompts actually engage the refusal circuit, so harm-vs-benign divergence is half-dominated by chat-style effects rather than refusal-specific ones. Raw-0-shot $\Delta \mathcal{L}_{\text{cap}}$ is small on

Table 9: Cross-model summary across three instruct-tuned 7 to 8B models. All replicate the suppression regime; the absolute q_{norm} band is family-specific; the behavioral attack number is a joint measurement of (refusal installed) $\times$ (robustness of that install). Geometry only; the behavioral effect of the attack is Table 1 in the main text, measured in a single run under both detectors.

Signal	Qwen-2.5-7B-Instruct	Meta-Llama-3-8B-Instruct	Mistral-7B-Instruct-v0.3
mean ρ	1.2×10^{-5}	5.6×10^{-6}	1.2×10^{-5}
peak $\rho/(k/d)$	$66\times$	$21\times$	$44\times$
peak $q_{\text{norm}}^{\text{REASON}}$ (layer)	220.6 ($\ell=19$)	48.9 ($\ell=30$)	158.7 ($\ell=12$)
peak $q_{\text{norm}}^{\text{ARITH}}$ (layer)	198.4 ($\ell=26$)	47.7 ($\ell=30$)	92.1 ($\ell=4$)
peak selective divergence	+0.166 ($\ell=24$)	+0.053 ($\ell=12$)	+0.032 ($\ell=13$)
max raw-0-shot $\Delta\mathcal{L}_{\text{cap}}$	+3.45	+0.88	+0.59

Llama (REASON+0.20, ARITH+0.88), well under the +1.1 3-shot-neutral ceiling; Llama-3-Instruct is less chat-template-dependent than Qwen.

Mistral-7B-Instruct-v0.3. $\rho \in [9 \times 10^{-6}, 3.2 \times 10^{-5}]$, 15 to $40\times$ the random floor, mean $\rho = 1.2 \times 10^{-5}$. Peak- q_{norm} band is early-to-mid residual: $q_{\text{norm}}^{\text{REASON}}$ peaks at 158.7 at $\ell = 12$; $q_{\text{norm}}^{\text{ARITH}}$ at 92.1 at $\ell = 4$ (see Fig. 5, right column). All three model families land in the suppression regime, but the absolute layer of the sharp q_{norm} band is family-specific. Behaviorally, Mistral-7B-Instruct-v0.3 is an outlier: 1.2% pre-attack refusal on AdvBench, so the Qi benign-fine-tuning attack has nothing to break. Post-attack refusal is 0.4% (post-attack ASR = 99.6% trivially). Compliance NLL *rises* from 5.01 to 6.13: benign Alpaca SFT pushes Mistral-Instruct further from a naked “Sure, here is” continuation. The attack number on Mistral measures install-thinness rather than install-fragility.

What the base already refuses. $\Delta = W_{\text{safe}} - W_{\text{base}}$ reads as a safety *install* only to the extent that W_{base} refuses nothing, so we measure the three shipped bases directly under the same prompts and detectors as their Instruct counterparts (no chat template is applied to either, so the pair is matched). Llama and Mistral behave as assumed, at or below 16.4% everywhere. Qwen does not: its base already refuses 71.1% of AdvBench by regex and 69.5% by classifier, so its shipped Δ raises AdvBench refusal by about 22 pp rather than creating it. This does not affect the geometry, which characterizes whatever Δ is, but it does mean the Qwen pair should be read as a partial install, and it is why the safety-only controls of Sec. 5 carry the weight they do. It also sharpens the fragility result: the attack leaves Qwen-Instruct at 58.2% on AdvBench (Table 1), *below* the 71.1% its own base refuses, so 100 benign steps do not merely undo the install but leave the model more compliant than the checkpoint it was built from.

Table 10: Refusal (%) of the three shipped *base* models, before any install, under both detectors. Same prompts, decoding and scoring as Table 1, with no chat template applied to either half of a pair. $n = 256/200/256$.

Base model	AdvBench		HarmBench		StrongReject	
	regex	classifier	regex	classifier	regex	classifier
Qwen-2.5-7B	71.1	69.5	31.5	34.0	28.5	30.9
Llama-3-8B	0.0	7.0	1.0	10.5	1.6	12.5
Mistral-7B-v0.3	0.0	16.4	0.0	14.0	0.8	19.9

F CHAT-TEMPLATE ABLATION: FULL FOUR-CONDITION TABLE

Scoring convention. Each probe defines a gold span that we score the NLL of, conditional on the prompt. For REASON (MMLU *formal_logic*) the gold span is a single letter (" A", " B", " C", or " D"). For ARITH (GSM8K) it is the numeric answer as a suffix. For RECALL (TriviaQA) it is the answer span. For INSTRUCT (FLAN-v2) it is the gold completion. This is the

same NLL-of-gold-span convention used by `lm-evaluation-harness` and HELM for MMLU-style benchmarks; the *scoring* is standard.

Why $\Delta\mathcal{L}_{\text{cap}}$ can inflate across a base/instruct pair. What is not automatically standard is using L_{cap} as a *capability-preservation diagnostic across two differently-aligned models*. The value L_{cap} is sensitive to how each model emits the first post-prompt tokens. A base model trained on raw-completion corpora tends to continue "...Answer:" with the bare letter directly, since that matches its training distribution. An instruct model trained with chat-style responses often continues with "The answer is A, because..." or an explanatory preamble that puts a newline or word before the gold letter. The probability mass on " A" drops, L_{cap} rises, and ΔL_{cap} looks large. The underlying knowledge is unchanged; only the output distribution’s *format* has shifted.

Ablation design. We run each probe under four prompt formats on Qwen-2.5-7B and Qwen-2.5-7B-Instruct. *raw (0-shot)*: the prompt ends at “Answer:” and we score the gold span directly. *raw (3-shot)*: three worked examples precede the prompt (in Q/A pairs), then the query, following a standard few-shot convention. *chat (0-shot)*: the prompt is wrapped in Qwen’s chat template. *chat (3-shot)*: chat template plus three in-context demonstrations. Table 11 reports $\Delta L_{\text{cap}} = L_{\text{cap}}(W_{\text{safe}}) - L_{\text{cap}}(W_{\text{base}})$ per condition.

Table 11: Chat-template ablation on Qwen-2.5-7B. Four prompt formats on the same four capability probes, $n = 64$, few-shot $k = 3$. Raw 0-shot matches Table 8 within sampling noise. 3-shot collapses the gap by 40 to 74% by showing the Instruct model the raw-completion format via in-context examples. Chat formatting *inflates* the gap because we score NLL of a bare gold token that Instruct no longer naturally emits after its chat template; 3-shot partially recovers by demonstrating the target format in context.

Probe	raw (0-shot)	raw (3-shot)	chat (0-shot)	chat (3-shot)
Reason	+3.35	+0.87	+21.01	+11.07
Arith	+3.06	+1.10	+15.19	+9.14
Recall	+0.90	+0.54	+5.39	+0.84
Instruct	+0.46	+0.37	+3.16	+2.99

Reading the table. The raw-0-shot column (first data column) reproduces the $\Delta\mathcal{L}_{\text{cap}}$ values in Table 8 within sampling noise. The raw-3-shot column shows the gap collapse to +0.37 to +1.10 nats across probes; this is our “format-neutralized” estimate of the true capability-loss shift and is small enough to be consistent with the geometric claim $\rho \ll \rho_{\text{part}}$. The chat columns go in the opposite direction: with the Instruct model’s expected chat template applied, the NLL of the bare gold token *rises* further, because the Instruct model even more strongly prefers to emit a preamble under its native format. This inversion confirms that the raw-0-shot $\Delta\mathcal{L}_{\text{cap}}$ in Table 8 is dominated by a format-shift effect rather than a capability shift.

G ERASURE IS REACHABLE: ADVERSARIAL-SFT DEMONSTRATION

None of the safety-training recipes studied in the main body crosses ρ_{part} . To establish that the erasure regime is not a definition without referents, we run a stress test that engineers an update which should, by construction, land inside F_{cap} ’s top- k eigenspace: supervised fine-tuning on (prompt, wrong-answer) pairs drawn from the capability probe itself. The gradient of NLL on a wrong gold token lies in the same directions the probe’s Fisher picks out, so in-subspace mass should accumulate with training steps. This is a deliberately contrived recipe and not a realistic alignment procedure; its purpose is a boundary-reachability check.

Setup. Base model: Pythia-1.4B `step143000`, matching §4. Two target probes are studied in parallel, one per independent adversarial-SFT run:

- *Target*=REASON: MMLU `formal_logic`, gold letter replaced by a uniformly-sampled non-gold letter. 2000 pairs, 2 epochs, LR 2×10^{-5} , AdamW, bf16.

- *Target=ARITH*: GSM8K training split, numeric gold answer perturbed by a small random offset preserving magnitude. Same 2000/2/2e−5 schedule.

The geometry instrument of §3 is then applied to each resulting W_{safe} against the Pythia-1.4B base: $N = 128$ per-example gradients, $k = 128$ per-layer subspace, evaluated on all four capability probes (target and three off-target).

Result: target-probe ρ reaches the ρ_{part} boundary. Table 12 summarizes. On the target probe, max ρ reaches 1.04×10^{-2} (target=REASON), at ρ_{part} , and 3.01×10^{-3} (target=ARITH). Mean ρ on target is 3.33×10^{-3} and 1.16×10^{-3} respectively, 10 to 30 $\times$ the SFT+DPO sub-pattern (b) values in §6. q_{norm} on target reaches 33,945 (reason) and 15,064 (arith), more than an order of magnitude above the pre-substrate OLMo $q_{\text{norm}}^{\text{ARITH}}$ of 863. The $\rho_{\text{part}} = 10^{-2}$ threshold is thus empirically anchored: it lies exactly where an update built to saturate F_{cap} ’s top- k in fact reaches.

target probe	probe under eval	mean ρ	max ρ	max q_{norm}	$\Delta\mathcal{L}_{\text{cap}}$	target?
REASON	REASON	3.33×10^{-3}	1.04×10^{-2}	33945	−0.96	✓
REASON	ARITH	1.16×10^{-3}	4.66×10^{-3}	15773	−0.62	
REASON	RECALL	4.97×10^{-4}	1.49×10^{-3}	1895	−0.35	
REASON	INSTRUCT	4.12×10^{-4}	1.33×10^{-3}	328	+0.15	
ARITH	ARITH	1.16×10^{-3}	3.01×10^{-3}	15064	−2.08	✓
ARITH	REASON	3.47×10^{-4}	8.95×10^{-4}	2956	−0.72	
ARITH	RECALL	2.78×10^{-4}	8.28×10^{-4}	1151	−0.38	
ARITH	INSTRUCT	1.84×10^{-4}	8.18×10^{-4}	316	+0.23	

Table 12: Adversarial-SFT erasure demonstration on Pythia-1.4B step143000. One SFT run per target probe; geometry measured against the base on all four probes. Max ρ on the target probe reaches the $\rho_{\text{part}} = 10^{-2}$ boundary for target=REASON and one order below for target=ARITH; q_{norm} on target is 10^2 to $10^3 \times$ any value seen under DPO/RLHF. Off-target probes see 3 to 8 $\times$ lower mean ρ than the target, establishing target specificity. $\Delta\mathcal{L}_{\text{cap}}$ is NLL-based; on capability probes with a stable wrong-token format (REASON, ARITH), adversarial SFT *lowers* NLL because format-learning dominates at this intervention scale (see caveat below). $\rho_{\text{rand}} = 3.75 \times 10^{-6}$ throughout.

Target specificity. Off-target probes see material but consistently lower ρ than the target probe: target=REASON elevates ARITH to 1.16×10^{-3} (2.9 $\times$ below target REASON) and RECALL/INSTRUCT near 4 to 5×10^{-4} (6 to 8 $\times$ below). Target=ARITH elevates REASON to 3.47×10^{-4} , RECALL to 2.78×10^{-4} . The ratio $\rho^{\text{target}}/\rho^{\text{off-target}}$ is stable across both runs, supporting a probe-local interpretation of F_{cap} ’s top- k : the eigendirections carrying reasoning and arithmetic overlap but are not identical, and an update engineered for one is not trivially an update for the other. This matters for downstream unlearning: an adversarial-SFT-style erasure on one skill is not a proxy for blanket capability removal.

NLL-vs-argmax caveat. The $\Delta\mathcal{L}_{\text{cap}}$ column shows that NLL on the target probe *decreases* after adversarial SFT (−0.96 on REASON, −2.08 on ARITH), the opposite of what a literal capability-loss reading would suggest. This is the format-learning effect: a Pythia-1.4B base emits letter and digit tokens after “Answer:” with nontrivial entropy, and 2000 demonstrations of *any* single-token answer (correct or not) sharpen the format distribution enough to lower NLL on the bare gold token. The intervention sharpens the format while corrupting content; argmax accuracy would drop where NLL improves. For consistency with the geometry machinery elsewhere in the paper we report NLL here, but the behavioral interpretation should be: adversarial SFT installs a large in-subspace perturbation ($\rho \rightarrow \rho_{\text{part}}$), the lemma’s erasure prediction applies (first-order capability change is permitted; the sign depends on what was optimized), and the NLL metric becomes unreliable because it does not factor format from content. A cleaner probe for future work is top-1 accuracy on held-out target-probe questions.

Interpretation. The result closes the boundary-reachability question without claiming that adversarial SFT is a practical unlearning method: it is not; it is a deliberately over-specified recipe designed to populate the erasure regime empirically. The takeaway for the main text is structural:

$\rho_{\text{part}} = 10^{-2}$ is not an arbitrary threshold but the value that an update built to saturate F_{cap} 's top- k eigenspace at this scale in fact reaches, and every suppression observation in the paper sits at least $50\times$ below it. The qualitative regime separation between suppression and erasure is therefore well-anchored on both sides.

H FULL PROOF OF LEMMA 1 (KERNEL IMMOBILITY)

We restate the lemma and give a full proof.

Lemma 2 (Kernel immobility, restated). *Let $\mathcal{L}_{\text{cap}}^{(c)} : \mathbb{R}^d \rightarrow \mathbb{R}$ be a capability loss, twice continuously differentiable on a neighborhood $\mathcal{N}$ of W_{base} , with Lipschitz Hessian on $\mathcal{N}$. Let*

$$F_{\text{cap}}^{(c)} := \mathbb{E}_{(x,y) \sim \mathcal{D}_c} [\nabla_{\theta} \log p_{\theta}(y|x) \nabla_{\theta} \log p_{\theta}(y|x)^{\top}] \Big|_{\theta=W_{\text{base}}}$$

be the empirical Fisher information matrix. Let $E := H(W_{\text{base}}) - F_{\text{cap}}^{(c)}$ denote the Gauss-Newton residual. If $\Delta \in \ker(F_{\text{cap}}^{(c)})$, then

$$\mathcal{L}_{\text{cap}}^{(c)}(W_{\text{base}} + \Delta) - \mathcal{L}_{\text{cap}}^{(c)}(W_{\text{base}}) = \frac{1}{2} \Delta^{\top} E \Delta + O(\|\Delta\|_2^3).$$

Under exact Gauss-Newton ($E = 0$, e.g., at a well-specified mature checkpoint where $p_{\theta}(\cdot | W_{\text{base}})$ matches $\mathcal{D}_c$ (Martens, 2020)), this strengthens to $O(\|\Delta\|_2^3)$. Either way, no first-order post-hoc update can reduce $\mathcal{L}_{\text{cap}}^{(c)}$ at leading order.

Proof. Step 1 (Taylor expansion). By Taylor's theorem with Lagrange remainder, for any Δ with $W_{\text{base}} + \Delta \in \mathcal{N}$,

$$\mathcal{L}_{\text{cap}}^{(c)}(W_{\text{base}} + \Delta) = \mathcal{L}_{\text{cap}}^{(c)}(W_{\text{base}}) + \nabla \mathcal{L}_{\text{cap}}^{(c)}(W_{\text{base}})^{\top} \Delta + \frac{1}{2} \Delta^{\top} H(W_{\text{base}}) \Delta + R(\Delta),$$

where $H(W_{\text{base}}) = \nabla^2 \mathcal{L}_{\text{cap}}^{(c)}(W_{\text{base}})$ is the Hessian and the remainder satisfies $|R(\Delta)| \leq \frac{L}{6} \|\Delta\|_2^3$ by the Lipschitz-Hessian assumption.

Step 2 (linear term vanishes from the kernel condition). The kernel hypothesis $\Delta \in \ker(F_{\text{cap}}^{(c)})$ gives $\Delta^{\top} F_{\text{cap}}^{(c)} \Delta = \mathbb{E}_{(x,y) \sim \mathcal{D}_c} [(\Delta^{\top} \nabla \log p_{\theta}(y|x))^2] = 0$. As an expectation of a nonnegative quantity, this forces $\Delta^{\top} \nabla \log p_{\theta}(y|x) = 0$ for $\mathcal{D}_c$ -almost every (x, y) . Taking expectations again,

$$\Delta^{\top} \nabla \mathcal{L}_{\text{cap}}^{(c)}(W_{\text{base}}) = -\Delta^{\top} \mathbb{E}_{(x,y) \sim \mathcal{D}_c} [\nabla \log p_{\theta}(y|x)] = 0.$$

The linear Taylor term thus vanishes from the kernel condition alone, irrespective of whether W_{base} is a stationary point of $\mathcal{L}_{\text{cap}}^{(c)}$.

Step 3 (Hessian, Fisher, and the Gauss-Newton residual). For the likelihood loss $\mathcal{L}_{\text{cap}}^{(c)} = -\mathbb{E}_{(x,y) \sim \mathcal{D}_c} [\log p_{\theta}(y|x)]$, the calculus identity $\nabla^2 \log p = \nabla^2 p/p - \nabla \log p (\nabla \log p)^{\top}$ gives

$$H(W_{\text{base}}) = -\mathbb{E}[\nabla^2 \log p] = \mathbb{E}[\nabla \log p \nabla \log p^{\top}] - \mathbb{E}[\nabla^2 p/p] = F_{\text{cap}}^{(c)} - \mathbb{E}[\nabla^2 p/p].$$

Bartlett's second identity gives $\mathbb{E}_{p_{\theta}}[\nabla^2 p/p] = \int \nabla^2 p \, dy = \nabla^2 \int p \, dy = 0$ when expectations are taken under the model's own output distribution. When $p_{\theta}(\cdot | W_{\text{base}})$ is well-specified for $\mathcal{D}_c$ (which holds at a well-trained mature checkpoint), the Gauss-Newton residual $E := H(W_{\text{base}}) - F_{\text{cap}}^{(c)} = -\mathbb{E}_{\mathcal{D}_c}[\nabla^2 p/p]$ vanishes; this is the standard Gauss-Newton approximation (Martens, 2020). In general E is a finite matrix.

Step 4 (combining). Substituting Step 2 into Step 1 and decomposing $H = F_{\text{cap}}^{(c)} + E$,

$$\mathcal{L}_{\text{cap}}^{(c)}(W_{\text{base}} + \Delta) - \mathcal{L}_{\text{cap}}^{(c)}(W_{\text{base}}) = \frac{1}{2} \Delta^{\top} F_{\text{cap}}^{(c)} \Delta + \frac{1}{2} \Delta^{\top} E \Delta + R(\Delta).$$

The kernel hypothesis kills the first term ($\Delta^{\top} F_{\text{cap}}^{(c)} \Delta = 0$). The Lipschitz-Hessian assumption gives $|R(\Delta)| \leq \frac{L}{6} \|\Delta\|_2^3$, so

$$\mathcal{L}_{\text{cap}}^{(c)}(W_{\text{base}} + \Delta) - \mathcal{L}_{\text{cap}}^{(c)}(W_{\text{base}}) = \frac{1}{2} \Delta^{\top} E \Delta + O(\|\Delta\|_2^3) = O(\|E\|_2 \cdot \|\Delta\|_2^2) + O(\|\Delta\|_2^3).$$

Under exact Gauss-Newton ($E = 0$), this collapses to $O(\|\Delta\|_2^3)$.

Step 5 (first-order corollary). A first-order optimizer (SGD, Adam, AdamW) minimizing an auxiliary objective $\mathcal{L}_{\text{safe}}$ produces an update of the form $\Delta = -\eta M \nabla \mathcal{L}_{\text{safe}} + O(\eta^2)$, where M is a (possibly diagonal, possibly identity) preconditioner. The lemma’s premise is on Δ itself, not on $\nabla \mathcal{L}_{\text{safe}}$ directly, so preconditioning does not weaken the conclusion: the lemma applies whenever the resulting Δ satisfies $\Delta \in \ker(F_{\text{cap}}^{(c)})$, which is the condition we measure empirically through $\rho^{(c)} = \|\Delta\|_2^2 / \|\Delta\|_2^2$. $\square$

Remark. The kernel condition $\Delta \in \ker(F_{\text{cap}}^{(c)})$ does the work in Step 2 alone: the linear Taylor term vanishes regardless of whether W_{base} is a stationary point of $\mathcal{L}_{\text{cap}}^{(c)}$. The local-minimum / well-specification assumption enters only at Step 3, where it underpins the Gauss-Newton approximation $H(W_{\text{base}}) \approx F_{\text{cap}}^{(c)}$. Under-trained checkpoints (e.g., the pre-substrate revisions in Sec. 6) violate well-specification: $\|E\|$ is large there, the strong $O(\|\Delta\|^3)$ form fails, and $\frac{1}{2} \Delta^\top E \Delta$ contributes a measurable $O(\|\Delta\|^2)$ shift to $\mathcal{L}_{\text{cap}}$, which is exactly the under-trained-base sub-pattern observed empirically. The post-substrate mature checkpoint where gradient optimization has converged is where $\|E\| \approx 0$ and the cubic bound applies; this is the regime the paper’s headline claim sits in.

I DEVELOPMENTAL-AXIS FOLLOW-UP: ALIGNMENT, CROSS-FAMILY GRAD-NORM, PLATEAU NOISE

We report three follow-up checks tied to claims in §6 and §5: (i) the empirical alignment $\cos(\nabla \mathcal{L}_{\text{cap}}, \Delta)$ at OLMo-2-1B revisions where Sec. 6’s SFT+DPO produced a usable W_{safe} ; (ii) the gradient-norm instrument applied to Pythia-1.4B revisions (cross-family lemma sanity check); (iii) a multi-seed re-run of the substrate-divergence sweep to bound plateau noise. Table 13 first gives the full per-revision substrate sweep behind Fig. 3.

Table 13: Substrate sweep on OLMo-2-0425-1B (full data behind Fig. 3): W_{base} swept across nine pretraining revisions spanning 0 to 4T tokens, $W_{\text{safe}} = \text{OLMo-2-0425-1B-Instruct}$ fixed. Random-subspace floor for this model is $k/d \approx 2.6 \times 10^{-6}$. Selective divergence transitions from noise to plateau between 6B and 60B pretraining tokens. ℓ^* = argmax-over-layers of selective divergence; reported as n/a at pre-substrate checkpoints where the curve is flat at noise ($\sim 10^{-4}$) and argmax would label a random fluctuation rather than a locus.

Step	Tokens	mean ρ	max selective_div $_{\ell}$	ℓ^*
0	0B	1.4×10^{-6}	$+4 \times 10^{-4}$	n/a
300	1B	1.6×10^{-6}	$+4 \times 10^{-4}$	n/a
1×10^4	21B	1.8×10^{-6}	$+2.7 \times 10^{-2}$	16
3×10^4	63B	1.3×10^{-6}	$+7.3 \times 10^{-2}$	16
5×10^4	105B	1.0×10^{-6}	$+7.7 \times 10^{-2}$	6
2×10^5	420B	9.3×10^{-7}	$+7.8 \times 10^{-2}$	9
5×10^5	1049B	9.1×10^{-7}	$+4.9 \times 10^{-2}$	9
1×10^6	2098B	8.3×10^{-7}	$+5.9 \times 10^{-2}$	9
1.9×10^6	4001B	3.4×10^{-6}	$+7.1 \times 10^{-2}$	11

Gradient-orthogonality at the OLMo developmental axis. Table 14 reports $\cos(\nabla \mathcal{L}_{\text{cap}}^{(c)}(W_{\text{base}}), \Delta)$ averaged across the four probes, where $\Delta = W_{\text{safe}} - W_{\text{base}}$ is the SFT+DPO update at each revision (Sec. 6). The relevant first-order capability-loss change per unit $\|\Delta\|^2$ is the linear coefficient $(\nabla \mathcal{L}_{\text{cap}})^{\top} \Delta / \|\Delta\|^2$.

Pythia-1.4B grad-norm cross-family replication. The U-shape in $\|\nabla \mathcal{L}_{\text{cap}}\|$ across OLMo-2-1B (Sec. 6: large at 1B and 4T tokens, small in 84 to 210B) might have been specific to OLMo-2-1B’s $\sim 200\times$ -Chinchilla overtraining ratio. We re-apply the gradient-norm instrument across eight Pythia-1.4B pretraining revisions (Table 15). *The U-shape replicates:* $\|\nabla \mathcal{L}_{\text{cap}}\|$ falls from 119 at step512 to a minimum of 28.8 at step30000 (~ 63 B tokens, ~ 45 tokens/param, near-Chinchilla), then

Table 14: Alignment of the safety update with the capability gradient at OLMo developmental-axis revisions. Mean is over the four capability probes; Δ is the SFT+DPO update from Sec. 6. Across the entire axis $|\cos(\nabla\mathcal{L}_{\text{cap}}, \Delta)| < 0.002$, including the over-trained 4001B endpoint where $\|\nabla\mathcal{L}_{\text{cap}}\|$ peaks at 29.6. Translated into absolute nats, the linear Taylor term $(\nabla\mathcal{L}_{\text{cap}})^\top \Delta = \cos\|\nabla\mathcal{L}_{\text{cap}}\| \cdot \|\Delta\|$ is at most ~ 0.11 nats at the pre-substrate step300 revision and ≤ 0.06 nats at post-substrate mature revisions: a fraction of 10^{-3} of the maximum-aligned upper bound $\|\nabla\mathcal{L}_{\text{cap}}\| \cdot \|\Delta\| \in [66, 240]$, confirming the kernel-induced gradient orthogonality empirically.

Revision	$\ \Delta\ $	mean $\ \nabla\mathcal{L}_{\text{cap}}\ $	mean $\cos(\nabla\mathcal{L}_{\text{cap}}, \Delta)$	mean linear/ $\ \Delta\ ^2$
step300-tokens1B	12.69	20.46	-6×10^{-4}	-7×10^{-4}
step10000-tokens21B	9.65	13.69	-5×10^{-4}	-6×10^{-4}
step20000-tokens42B	9.11	10.09	-1×10^{-4}	-3×10^{-5}
step50000-tokens105B	8.87	7.97	0 to 4 d.p.	$+3 \times 10^{-4}$
step1907359-tokens4001B	8.12	29.56	-4×10^{-4}	-6×10^{-4}

rises again as Pythia is trained past compute-optimal, reaching 50.5 at the final step143000 (~ 100 tokens/param, $\sim 5\times$ Chinchilla). This argues against an OLMo-specific overtraining artifact; the U-shape reflects a generic property of pretraining where weight-decay-driven non-stationarity grows on either side of the compute-optimal point.

Table 15: $\|\nabla\mathcal{L}_{\text{cap}}\|$ across Pythia-1.4B pretraining revisions, mean over the four capability probes. The U-shape from OLMo replicates: minimum at step30000 (~ 63 B tokens, near-Chinchilla), large at both extremes.

Revision	Tokens	mean $\ \nabla\mathcal{L}_{\text{cap}}\ $	REASON	ARITH	RECALL	INSTRUCT
step512	1B	119.5	146.7	209.9	98.7	22.5
step1000	2B	96.0	100.5	140.3	128.5	14.9
step3000	6B	38.8	51.5	72.2	22.3	9.1
step10000	21B	32.6	50.4	51.5	21.0	7.5
step30000	63B	28.8	31.2	56.1	19.8	8.2
step60000	126B	32.7	41.4	59.3	22.4	7.8
step100000	210B	41.8	51.0	75.1	30.5	10.5
step143000	300B	50.5	63.0	91.4	35.0	12.8

Substrate-plateau noise floor. The single-seed substrate sweep (§6) showed apparent within-plateau bouncing ($\text{max selective_div}_\ell \in [0.04, 0.12]$ across 42 to 210B; $\text{argmax layer} \in \{6, 7, 8, 9, 16\}$) at a single deterministic prompt order ($n_{\text{pairs}}=50$ from a pool of 60). We re-run with three shuffled seeds per revision (Table 16). Within-revision standard deviation is 0.002 to 0.010, an order of magnitude smaller than the between-revision spread of ~ 0.08 . Argmax layer is stable across seeds (one-position drift only at 168B, 210B). The two peaks at 84B and 147B are well above 1σ from their neighbors, they are real structure, not noise. The corrected reading: substrate engagement begins at ~ 42 B, reaches a first peak at 84B, dips moderately at 126B, peaks again at 147B, and stabilizes at ~ 0.07 from 168B onward.

J PER-REVISION DPO: FULL THREE-RECIPE TABLE

Protocol. We apply three DPO recipes of increasing compute to the same OLMo-2-0425-1B revisions straddling the substrate transition. *Weak*: HH-RLHF 5K pairs, 1 epoch, $\beta = 0.1$ (the §4 recipe) at 5 revisions. *Strong*: HH-RLHF 20K pairs, 2 epochs at 5 revisions. *SFT+DPO*: a two-stage pipeline that SFTs on 20K Tulu-3-SFT-Mixture (Lambert et al., 2024) examples then DPOs on 20K HH-RLHF pairs for 2 epochs, at 5 revisions, a reduced-budget approximation of the shipped OLMo-2-Instruct recipe (omitting the RLVR stage). All three measure $\rho^{(c)}(t)$, $q_{\text{norm}}^{(c)}(t)$, selective divergence, and pre-/post-attack refusal per matched pair.

Table 16: Substrate divergence at 3 shuffled-seed runs per revision (mean $\pm$ std). Argmax layers reported per-seed, in run order.

Revision	max selective_div $_{\ell}$	argmax layer (per seed)
step20000-tokens42B	0.043 $\pm$ 0.002	16, 16, 16
step40000-tokens84B	0.100 $\pm$ 0.008	9, 9, 9
step60000-tokens126B	0.073 $\pm$ 0.007	6, 6, 6
step70000-tokens147B	0.122 $\pm$ 0.009	6, 6, 6
step80000-tokens168B	0.068 $\pm$ 0.010	7, 7, 6
step90000-tokens189B	0.063 $\pm$ 0.005	8, 8, 8
step100000-tokens210B	0.073 $\pm$ 0.008	7, 7, 6

Table 17: Per-revision DPO on OLMo-2-0425-1B under three recipes of increasing strength. $q_{\text{norm}}^{\text{ARITH}}$ is the per-recipe layer-max on the ARITH probe. All rows are in the suppression regime by absolute ρ . Weak/Strong show the under-trained-base sub-pattern: pre-substrate $q_{\text{norm}}^{\text{ARITH}}$ (620/863 at 1B) is more than an order of magnitude above any of their post-substrate values (13 to 48). SFT+DPO shows the SFT-stage-recipe sub-pattern instead: ρ stays elevated ($\sim 10^{-4}$) and $q_{\text{norm}}^{\text{ARITH}}$ stays high (95 to 770) at every revision, reflecting the SFT-stage NLL gradient projecting onto F_{cap} ’s top- k regardless of substrate maturity. Behavioral refusal appears only under SFT+DPO; Weak/Strong produce no measurable refusal at any revision (behavioral columns omitted). This is a weight-space geometry table, not the fragility result: the headline fragility claim is on Qwen/Llama (§4), and the one developmental behavioral point that exists, SFT+DPO at 4001B, does itself show a successful attack (post-attack refusal 15.6%, ASR 84.4%), so the table should not be read as “no successful attack.” Source for the prose summary in §6.

Step	Tokens	Weak		Strong		SFT+DPO			
		ρ	$q_{\text{norm}}^{\text{ARITH}}$	ρ	$q_{\text{norm}}^{\text{ARITH}}$	ρ	$q_{\text{norm}}^{\text{ARITH}}$	pre-ref	post-ref / ASR
3×10^2	1B	8.1×10^{-5}	620	9.5×10^{-5}	863	8.3×10^{-5}	95	21.9%	0% / 100%
3×10^4	63B	4.6×10^{-6}	23	6.8×10^{-6}	34	5.8×10^{-5}	603	0%	0% / 100%
5×10^4	105B	4.3×10^{-6}	13	6.6×10^{-6}	13	8.0×10^{-5}	283	4.7%	3.1% / 96.9%
5×10^5	1049B	7.2×10^{-6}	33	1.2×10^{-5}	40	1.1×10^{-4}	770	0%	0% / 100%
1.9×10^6	4001B	9.8×10^{-6}	22	1.8×10^{-5}	48	5.9×10^{-5}	529	42.2%	15.6% / 84.4%

K CO-TRAINING: FULL PROTOCOL AND PER-CONDITION TABLES

Sample sizes, and the 410M scale. Every table in this appendix reports the *original* from-scratch runs, evaluated on 64 AdvBench prompts. A separate, higher-powered rerun on 256 prompts exists for cotrain-continuous only, so both the schedule table (Table 2, main text) and the four-condition table below rest on the 64-prompt runs. The main-text scale table (Table 3) reports the rerun, and it is the run the body quotes throughout. At 410M the two runs read **84.4** $\rightarrow$ **84.4%** at $n = 64$ here and **98.4** $\rightarrow$ **84.0%** at $n = 256$ there. The post-attack values agree; the pre-attack values do not, and that single gap is the whole discrepancy between a 0 pp shift and a -14.4 pp one. The powered figure is the one to trust: the install erodes modestly rather than not at all, still well short of the 35 to 38 pp that post-hoc installs lose (Table 1).

Protocol (Pythia-410M from-scratch). Random-initialized Pythia-410M architecture trained on streaming allenai/c4 for $N \approx 10$ B tokens, sequence length 2048, micro-batch 8 per rank, global batch 512 sequences via gradient accumulation across 2 ranks (DDP). AdamW with peak lr 3×10^{-4} , $\beta = (0.9, 0.95)$, weight decay 0.1, gradient clipping 1.0, cosine schedule with 1% warmup. $\mathcal{L}_{\text{safe}}$ is cross-entropy on (AdvBench harmful prompt, refusal-target) pairs: each prompt is paired round-robin with one of 5 short refusal-template targets (“I can’t help with that.”, “As an AI assistant, I cannot ...”, etc.); during safety-active steps each step adds a microbatch of 2 safety pairs weighted by $\lambda(t)$. The training and evaluation prompt pools are both drawn from the public AdvBench CSV under different sampling seeds and partially overlap by construction (training uses 256 prompts from Random(42), evaluation uses 64 from Random(101)); this is a deliberate choice, the install we measure is whether the model reproduces *the refusal-template distribution* on harmful prompts, not whether it has memorized prompt-specific responses, so prompt overlap is not a leakage path for that quantity. The two co-training schedules differ only in $\lambda(t)$:

Table 18: From-scratch Pythia-410M, 10B C4 tokens, four conditions (referenced from Sec. 7). Refusal rate on 64 AdvBench prompts before and after the benign-SFT attack. This is the original run; the four-condition comparison is valid within it, but for `cotrain-continuous` the higher-powered 256-prompt rerun gives $98.4 \rightarrow 84.0$ rather than the 0 pp shift here, so the erosion is modest rather than absent. Capability is per-probe NLL, where *lower is better*: `cotrain-continuous` matches or improves on the LM-only base on every probe (e.g. arithmetic 8.6 versus the base’s 10.0, a small improvement, not a degradation). It is the only condition that installs refusal, and its install is fully attack-robust at matched-or-better capability.

Condition	Pre-ref	Post-ref	Shift	NLL _{arith}	NLL _{reason}
base	0.0%	0.0%	0 pp	10.01	5.34
cotrain-windowed	0.0%	1.6%	+1.6 pp	9.11	4.78
cotrain-continuous	84.4%	84.4%	0.0 pp	8.57	4.95
post-hoc-dpo	0.0%	0.0%	0 pp	10.00	5.45

- `cotrain-windowed`: $\lambda(t) = 0.5$ for $t \in [0.10, 0.30] \cdot T$, $\lambda(t) = 0$ otherwise.
- `cotrain-continuous`: $\lambda(t) = 0.5 \cdot (0.30 - 0.10) = 0.10$ for all t , equal total safety-loss compute to the windowed schedule.

`post-hoc-dpo` runs 5K-pair HH-RLHF DPO on the base output. Attack: 200-step / $1r \times 10^{-5}$ benign-SFT on 100 Alpaca examples, evaluated on 64 AdvBench prompts (same recipe as Sec. 4).

Geometry across conditions (Table 19). Each non-base condition’s geometry is measured against base as W_{base} . All three lie in the suppression regime; the (ρ, q_{norm}) instrument does not distinguish behaviorally robust from behaviorally fragile installs at this scale.

Table 19: Geometry of the safety direction at 410M scale, measured against `cond_base`. Mean over 96 (layer, probe) cells. All three conditions sit in the suppression band; behaviorally, only `cotrain-continuous` resists the benign-SFT attack. Refusal is the original 64-prompt run, so `cotrain-continuous`’s 84.4% here is the $n = 64$ counterpart of the 84.0% at $n = 256$ quoted above; the other two conditions have no 256-prompt rerun.

Condition	mean ρ	mean q_{norm}	post-attack refusal
cotrain-continuous (uniform)	2.77×10^{-5}	2.31	84.4%
cotrain-windowed	2.75×10^{-5}	2.27	1.6%
post-hoc-dpo	3.01×10^{-5}	4.01	0.0%

Where co-trained safety sits relative to capability (Table 20). The Δ -based instrument of Fig. 1a cannot place a co-trained model, for the reason given in Limitation (vii): no post-hoc displacement exists, and the only available Δ is a between-trajectory gap whose small ρ is close to a geometric tautology. The within-model instrument does apply: read the safety gradient inside a single model and project it onto that same model’s F_{cap} . Comparing `cotrain-continuous` against the LM-only base at the same scale gives a sign-consistent *double dissociation*. ρ_{safe} falls on REASON at all four scales (0.90, 0.68, 0.54, 0.74 $\times$) and rises on INSTRUCT at all four (3.65, 1.33, 6.54, 2.05 $\times$); RECALL and ARITH move in both directions and carry no consistent signal. Co-trained safety is therefore entangled with instruction-following specifically rather than with capability at large, which is what makes an instruction-tuning attack able to erode it at all while stopping short of removal. The instrument separates co-training from post-hoc, but it does not rank schedules *within* co-training: at 410M the robust `cotrain-continuous` and the failed `cotrain-windowed` differ by under 2% in q_{norm} (2.31 against 2.27, Table 19) while differing by 83 pp in refusal, so a geometric predictor of which schedule survives remains open.

160M continued-training pilot. We previously ran a smaller continued-training pilot at Pythia-160M scale (continued-training from the shipped checkpoint on streaming C4 for $N = 2000$

Table 20: $\rho_{\text{safe}} (\times 10^{-3})$, LM-only base $\rightarrow$ cotrain-continuous at the same scale, per capability probe. Measured *within* a single model (safety gradient projected onto that model’s own F_{cap}), so unlike Fig. 1a’s ρ it needs no post-hoc Δ and is comparable across the two arms. REASON falls at every scale and INSTRUCT rises at every scale; RECALL and ARITH are mixed. Absolute values are not comparable to Fig. 1a’s ρ or to ρ_{part} , which are calibrated for a displacement rather than a gradient; only the within-scale ratio between the two arms is meaningful.

Scale	REASON	RECALL	ARITH	INSTRUCT
410M	0.93 $\rightarrow$ 0.83	1.62 $\rightarrow$ 1.79	0.88 $\rightarrow$ 0.96	3.56 $\rightarrow$ 12.98
1B	1.53 $\rightarrow$ 1.04	2.48 $\rightarrow$ 1.98	1.86 $\rightarrow$ 1.98	5.81 $\rightarrow$ 7.72
2.8B	1.35 $\rightarrow$ 0.74	1.98 $\rightarrow$ 0.43	1.75 $\rightarrow$ 0.60	3.56 $\rightarrow$ 23.24
6.9B	2.97 $\rightarrow$ 2.19	3.81 $\rightarrow$ 5.20	3.79 $\rightarrow$ 3.03	7.05 $\rightarrow$ 14.48

AdamW steps at $\text{lr } 5 \times 10^{-5}$, batch size 4). The four-condition table is reproduced here for completeness; the headline pattern is the same: a 500-step early window (cotrain-early) is washed out, while a 500-step late window (cotrain-late) installs refusal that the attack *partially erodes*. The from-scratch 410M result above generalizes the same finding: short-window co-training does not produce attack-robust refusal regardless of *when* the window sits during training; only continuous co-training does.

Table 21: 160M continued-training pilot for completeness. cotrain-early’s safety signal is washed out by subsequent LM-only steps; cotrain-late installs refusal that the attack partially erodes; the two DPO-style conditions install no measurable refusal at 160M (the 5K-pair DPO recipe does not engage). Refusal on 64 AdvBench prompts; this pilot has no 256-prompt rerun and no main-text counterpart.

Condition	Pre-attack refusal	Post-attack refusal	Post-attack ASR	Refusal shift
baseline	0.0%	0.0%	100%	0.0 pp
cotrain-early	0.0%	1.6%	98.4%	+1.6 pp
cotrain-late	84.4%	15.6%	84.4%	−68.8 pp
post-hoc-dpo	0.0%	0.0%	100%	0.0 pp

L DETECTOR CROSS-CHECK: REFUSAL GATE VERSUS HARMFUL OUTPUT

The main text scores refusal with a regex over refusal-prefix templates. That is the quantity the geometry speaks to: Δ installs a refusal *gate*, and the suppression reading predicts that benign SFT perturbs the gate while leaving the underlying capability routed to the original output distribution. A regex detects whether the gate fires. It does not detect whether what follows is harmful.

Those two questions can come apart, so we rescore every generation behind Tables 1 and 3 with Llama-Guard-3-8B, a learned classifier of harmful *output*. Both detectors run on the same saved generations in a single pass, so the columns below are directly comparable. Table 10 already reports the shipped base models under both.

Table 22: Table 1 under both detectors: refusal (%) before $\rightarrow$ after the same 100-example benign-SFT attack on the three shipped instruct models. The regex columns are the main-text table. $n = 256/100/64$; the three cells that rise are within noise.

Model	AdvBench		HarmBench		StrongReject	
	regex	classifier	regex	classifier	regex	classifier
Qwen-2.5-7B	93.0 $\rightarrow$ 58.2	93.0 $\rightarrow$ 85.2	67.5 $\rightarrow$ 45.5	74.0 $\rightarrow$ 59.5	52.3 $\rightarrow$ 56.6	72.3 $\rightarrow$ 73.8
Llama-3-8B	49.2 $\rightarrow$ 10.9	39.5 $\rightarrow$ 27.7	46.5 $\rightarrow$ 6.5	49.0 $\rightarrow$ 28.0	46.5 $\rightarrow$ 7.8	53.9 $\rightarrow$ 32.4
Mistral-7B-v0.3	1.2 $\rightarrow$ 0.4	12.1 $\rightarrow$ 6.2	1.5 $\rightarrow$ 0.0	7.0 $\rightarrow$ 8.5	3.5 $\rightarrow$ 0.8	23.0 $\rightarrow$ 14.1

Table 23: Table 3 under both detectors: from-scratch cotrain-continuous refusal (%), pre $\rightarrow$ post benign-SFT attack, by scale and prompt distribution. The regex columns are the main-text table. $n = 256/100/64$; changes under ~ 5 pp are within noise.

Scale	AdvBench (in-dist.)		HarmBench (out)		StrongReject (out)	
	regex	classifier	regex	classifier	regex	classifier
410M	98.4 $\rightarrow$ 84.0	91.8 $\rightarrow$ 64.8	58.0 $\rightarrow$ 34.5	57.0 $\rightarrow$ 33.5	8.2 $\rightarrow$ 7.4	25.8 $\rightarrow$ 21.5
1B	87.1 $\rightarrow$ 84.8	83.2 $\rightarrow$ 63.7	56.0 $\rightarrow$ 32.0	54.0 $\rightarrow$ 25.0	7.8 $\rightarrow$ 8.2	23.0 $\rightarrow$ 20.3
2.8B	96.1 $\rightarrow$ 90.2	91.0 $\rightarrow$ 77.3	45.5 $\rightarrow$ 31.5	49.0 $\rightarrow$ 37.5	8.6 $\rightarrow$ 8.2	31.2 $\rightarrow$ 19.5
6.9B	96.9 $\rightarrow$ 90.6	97.3 $\rightarrow$ 86.7	32.5 $\rightarrow$ 39.0	33.0 $\rightarrow$ 41.5	9.0 $\rightarrow$ 10.9	29.7 $\rightarrow$ 19.5

What changes, and what does not. The classifier preserves every ordering the main text rests on. Post-hoc installs still lose far more of their refusal to the attack than co-trained ones do, co-training is still the only condition that installs refusal at all (cotrain-continuous 82.8% against base 4.7% on the schedule runs), and the coverage gap between AdvBench and the two out-of-distribution benchmarks still widens with scale.

What the classifier changes is the level, and it does so in both directions. It reads Mistral and Llama as refusing more than the regex does before the attack (Mistral 1.2 against 12.1% on AdvBench), because those models decline in wordings the refusal-prefix regex does not match. It reads the co-trained models as eroding more (cotrain-continuous 82.8% to 65.6% on the schedule runs), because after the attack their refusal templates sometimes survive as a prefix while the continuation drifts toward compliance. Both effects are properties of the readout, not of the geometry: the gate and the output are different quantities, and only the first is what a weight-space displacement can be expected to predict. We report the regex in the main text for that reason, and report the classifier here so the gap is visible.

M LIMITATIONS AND BROADER IMPACT

Limitations. (i) F_{cap} is approximated by its top- k eigenspace; tail eigendirections carry some capability mass and are unmeasured, so a weight update with small top- k projection may still traverse low-curvature but semantically important directions that our instrument misses. (ii) Refusal detection is regex-based (“I can’t”, “as an AI”), not a trained harmfulness classifier (Mazeika et al., 2024); ASR numbers are typically 5 to 15 pp more generous than classifier-scored runs, so our claims rest on relative ranking across models and conditions rather than absolute ASR magnitudes. (iii) The fragility attack is Qi et al.’s benign-fine-tuning variant, the weakest of the three variants in that paper, so our “attack succeeds” claim is conservative. We separately verify that the continuous co-training install is robust to stronger attacks (Sec. 7): a benign-SFT sweep over learning rate and steps leaves worst-case post-attack refusal at 75% and an identity-shift persona attack leaves it at 100%; only the harmful-demonstration variant, which needs Qi et al.’s released data, is untested. (iv) The developmental-axis behavioral curve is single-point: SFT+DPO produces pre-attack refusal only at the final pretraining checkpoint (Sec. 6); a per-revision curve would require a full Tulu-3 pipeline (SFT $\rightarrow$ DPO $\rightarrow$ RLVR) at each revision, out of compute budget. (v) The co-training result (Sec. 7) supports continuous-schedule pretraining-time co-training at 410M from-scratch scale, but the head-to-head “robustness at matched pre-attack refusal” comparison against post-hoc-dpo is not directly available because the 5K-pair DPO recipe installs no measurable refusal at this scale (0% pre-attack); a stronger post-hoc baseline (SFT+DPO or Tulu-3-style pipeline at 410M) is needed to make the protection differential quantitative. We verify the result from scratch to 6.9B (Table 3); beyond that it remains a compute-bounded conjecture, and the refusal it installs is in-distribution. (vi) The geometric instrument runs to 8B; frontier-scale (70B+) behavior is a conjecture the framework makes but does not verify. (vii) The Δ -based instrument does not apply to the co-training conditions, though a within-model measurement does (Sec. 7). It characterizes an *edit* $\Delta = W_{\text{safe}} - W_{\text{base}}$ applied to a frozen, converged base: $\|\Delta\| = 0.0015$ for the post-hoc DPO at 410M and 2.9 to 7.8 for the shipped 7 to 8B instruct deltas. A co-trained model was never a capable-but-unsafe base that received a patch, so no such Δ exists; the only available one is the gap between two complete from-scratch runs, $\|\Delta\| = 86.5$. The four conditions share a seed and data order and differ only by the $\lambda\mathcal{L}_{\text{safe}}$ term, so that gap is attributable to the safety loss, but it is four to

five orders larger than a real install because adding the term sent the run down a different trajectory: it carries the divergence of the language-modeling path, not a localized safety displacement. Since ρ is scale-free it can still be computed and comes out small ($\approx 3 \times 10^{-5}$), but in a high-dimensional space almost any direction has a small projection onto a k -dimensional subspace, so for a between-trajectory gap that number is close to a geometric tautology rather than evidence of a thin install. This is also why Lemma 1 does not bear on co-training: its premise is a post-hoc displacement from a converged base. Two measurements settle whether co-trained safety is genuinely entangled with capability, and neither differences two runs. The first, a safety gradient read *within* a single model and projected onto that model’s own F_{cap} , we now report (Sec. 7): co-training raises ρ_{safe} on the INSTRUCT probe at every scale while mostly lowering it on REASON, RECALL and ARITH, so the entanglement is with instruction-following rather than with capability at large. The second remains open: sweep attack strength and record capability alongside refusal, so that co-trained refusal falling only once capability falls with it would demonstrate the entanglement behaviorally.

What the instrument does not claim. Beyond the Ethics Statement, one scope limit is worth stating separately: the instrument flags when an alignment procedure has landed in the suppression regime, a fine-tuning-robustness red flag, not when a model has been made unsafe by any behavioral standard. The two are different questions, and only the first is measurable from weight-space geometry.

N EXTENDED RELATED WORK

Post-hoc alignment and its fragility. RLHF and its direct-preference descendants (Christiano et al., 2017; Ouyang et al., 2022; Rafailov et al., 2023) are the standard alignment recipe. Qi et al. (2024) show that only 10 to 100 fine-tuning examples (even benign ones) compromise the safety of aligned GPT-3.5 and Llama-2-chat; the benign-fine-tuning variant is our behavioral attack. Zhan et al. (2024) strip RLHF from GPT-4 with 340 examples and 95% ASR; Andriushchenko et al. (2025) reach 100% ASR on leading aligned models via simple adaptive attacks; Paulus et al. (2025) drive suffix search two orders of magnitude faster. Wei et al. (2023) identify competing-objectives and mismatched-generalization as root causes; Anil et al. (2024) push these to long-context regimes. Closest to our weight-space account: Qi et al. (2025) localize safety to the first few output tokens (“shallow alignment”); Wei et al. (2024) find $\sim 3\%$ of parameters and 2.5% of ranks carry the entire safety install; Peng et al. (2024) map a narrow “safety basin” in weight space around aligned checkpoints. Our contribution is the continuous-geometry account (Fisher-eigenspace regime classification) that these local-pruning / shallow-feature findings are special cases of. Mitigations span prompt-template defenses (Huang et al., 2024a), training-time regularization (Lyu et al., 2024), perturbation-aware alignment (Huang et al., 2024b), representation-level noise defenses (Rosati et al., 2024), and latent adversarial training (Sheshadri et al., 2024); all operate inside the suppression regime and thus face the same first-order-free-motion fragility our Lemma predicts.

Mechanistic interpretability of alignment. Arditi et al. (2024) show RLHF-installed refusal is mediated by a single residual-stream direction; Wollschläger et al. (2025) extend this to multi-directional concept cones, and ablation restores full compliance in both cases. Lee et al. (2024) reverse-engineer DPO on toxicity and argue it installs an overlay rather than removing learned features. Zou et al. (2023a); Panickssery et al. (2024) steer behavior via contrastive activation vectors. Templeton et al. (2024); Marks & Tegmark (2024); Lieberum et al. (2024); Marks et al. (2025) find high-level concepts linearly encoded in residual streams of frontier-class models and develop SAE / sparse-circuit tooling to isolate them causally. Hubinger et al. (2024); Greenblatt et al. (2024) provide complementary behavioral evidence that safety training does not erase the underlying capability. Our per-layer ρ and q_{norm} operationalize these activation-level intuitions at the weight level, making the “install vs. remove” distinction geometric and quantitative.

Loss-landscape geometry and Fisher methods. The empirical Fisher matrix F_{cap} is standard in natural-gradient (Amari, 1998; Martens, 2020) and continual-learning literatures, most notably EWC (Kirkpatrick et al., 2017). We use the top- k eigenspace (not the diagonal), which captures multi-parameter correlations that matter when Δ is structured. Grosse et al. (2023) scale influence-function Fisher-Hessian estimates to 52B-parameter models; our top- k Gram trick is a lightweight per-layer variant. Keskar et al. (2017) relate flat minima to generalization; Jacot et al. (2018) give

the fixed-kernel infinite-width limit; Ortiz-Jiménez et al. (2023) show task arithmetic becomes exact in the NTK-linearized tangent space, providing the geometric foundation we extend to DPO. Task arithmetic and model merging (Ilharco et al., 2023; Wortsman et al., 2022; Yadav et al., 2023) treat safety- and task-vectors as weight-space objects; Hu et al. (2022) establish the low-intrinsic-rank hypothesis for fine-tuning updates that underlies tractable eigenspace analysis at LLM scale. Our decomposition explains why safety vectors are approximately orthogonal to capability vectors in real-scale models.

Developmental phenomena. Achille et al. (2018) introduce critical learning periods in vision networks; Frankle et al. (2020) localize the most consequential training period to the first few epochs; Kleinman et al. (2024) show the same arises in deep linear networks. Constantinescu et al. (2025) transfer the biological critical-period analogy directly to language models and find catastrophic-forgetting windows concentrated in the first $\sim 10\%$ of continued pretraining; our 6B to 60B substrate transition in OLMo-2-1B sits in the same developmental bracket. LLM emergent abilities (Wei et al., 2022; Srivastava et al., 2023; Schaeffer et al., 2023) turn on sharply at scale- or data-dependent thresholds, and the quantization-model view (Michaud et al., 2024) describes them as discrete “quanta” acquired during training; Chinchilla-style scaling (Hoffmann et al., 2022) places our substrate-transition window well inside compute-optimal token budgets. We connect these to safety: not only do capabilities emerge at a specific pretraining time, but the substrate that lets *safety* bind to those capabilities has its own developmental window.

Safety signal during pretraining. Korbak et al. (2023) inject preference feedback into pretraining objectives; Bai et al. (2022b) train from constitutional principles; Zhou et al. (2023) argue alignment needs only a small high-quality SFT. Our co-training result (Sec. 7) is complementary: a windowed-vs-continuous comparison at matched total safety-loss compute shows that *persistence* of the safety signal across pretraining, not its timing, drives attack robustness.

Unlearning as contrast. Machine-unlearning methods attempt to genuinely remove capability rather than install a refusal gate. Li et al. (2024) introduce the WMDP benchmark and the RMU method, which randomizes representations for targeted harmful knowledge; Lynch et al. (2024) benchmark eight unlearning methods and find capability is often reliably re-extractable. The contrast sharpens our framing: unlearning targets the erasure regime ($\rho \gtrsim \rho_{\text{part}}$), DPO/RLHF target the suppression regime; neither in isolation solves the fragility problem, consistent with Lemma 1 (erasure requires motion *inside* the capability subspace, which suppression-regime DPO does not provide).

Open-weight substrates. Pythia (Biderman et al., 2023) pioneered intermediate-checkpoint releases; OLMo-2 (OLMo et al., 2025) ships 267 checkpoints plus a Tulu-3 (Lambert et al., 2024)-trained Instruct variant with real refusal behavior (the combination we need for the developmental axis). Cross-family replications use Qwen-2.5 (Yang et al., 2024), Meta-Llama-3 (Grattafiori et al., 2024), and Mistral-7B-v0.3.